Rexroth
Bosch Group

Hydraulic cylinder
Mill type

RE 17331/09.05 1/44
Replaces: 02.05

Series CDH1 / CGH1

Component series: 1X (Piston Ø 40 to 200 mm)
2X (Piston Ø 220 to 320 mm)
Nominal pressure 250 bar (25 MPa)


HA/4646/95

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Features

- 6 mounting styles
- Piston Ø: 40 to 320 mm
- Piston rod Ø: 28 to 220 mm
- Stroke lengths up to 6 m



Engineering software: IHC-Designer from Rexroth

Online www.boschrexroth.com/Rexroth-IHD

Download www.boschrexroth.com/business_units/bri/de/downloads/ihc

Technical data (for applications outside these parameters, please consult us!)

Standards:
Bosch Rexroth Standard; main dimensions such as piston Ø and piston rod Ø meet the requirements of DIN ISO 3320.

Nominal pressure: 250 bar
Static proof pressure: 375 bar
Higher operating pressures on request.

The specified operating pressures are only valid for applications with shock-free operation. If extreme loads occur, eg. as happens in high sequence cycles, the fixing elements and piston rod thread connections need to be designed for durability (fatigue strength).

Installation: Optional

Pressure fluid / pressure fluid temperature range:
HL, HLP, HFD-R: –20 °C to +80 °C
HFA: +5 °C to +55 °C
Water glycole HFC on request

Viscosity range: 2.8 to 380 mm^2/s

ISO cleanliness class
Maximum permissible degree of pressure fluid contamination is to ISO 4406 (c) class 20/18/15.

Stroke velocity: Up to 0.5 m/s (dependent on the connection ports), higher stroke velocities on request

Bleed screw as standard: Secured against unscrewing

Undercoat: The hydraulic cylinders are, as standard, painted (colour tone gentian blue RAL 5010) with a max. thickness of 80 µm. Other colour tones on request.

Acceptance: Each cylinder is tested to Bosch Rexroth standards.

Cylinders whose application data lies outsdie the stated values can be offered as a special version.

Cylinders with piston Ø > 320 mm are available on request as an ABS (**A**pplications **B**ased **S**tandardisation) cylinder.

For assembly, commissioning and maintenance of hydraulic cylinders the operating guidelines RE 07100-B have to be taken into account!

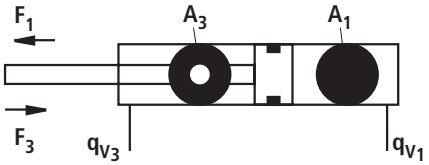
Diameter, weights

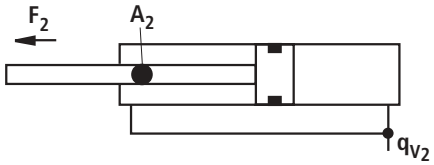
Piston	Piston rod	CD cylinder at 0 mm stroke length					Per 100 mm stroke length	CG cylinder at 0 mm stroke length			Per 100 mm stroke length
AL Ø mm	**MM** Ø mm	**MP3** [1] **MP5** [1] kg	**MP3** [2] **MP5** [2] kg	**MF3 MF4** kg	**MT4** kg	**MS2** kg	kg	**MF3** kg	**MT4** kg	**MS2** kg	kg
40	**22** **28**	7 7	12 12	9 9	9 9	9 9	0.9 1.0	10 10	9 9	9 10	1.2 1.5
50	**28** **36**	10 10	16.5 16.5	14 14	12 12	12 13	1.2 1.5	15 15	14 14	14 14	1.6 2.3
63	**36** **45**	16 16	25.5 25.5	22 22	19 19	19 20	2.1 2.6	24 24	21 22	21 22	2.9 3.8
80	**45** **56**	25 26	35 36	30 31	29 30	31 32	2.9 3.6	34 35	33 34	35 36	4.1 5.5
100	**56** **70**	43 44	58.5 59.5	52 53	50 51	52 53	4.6 5.7	59 60	56 58	58 60	6.6 8.8
125	**70** **90**	79 80	99 100	93 95	91 93	90 92	7.3 9.2	103 106	101 105	100 104	10.3 14.2
140	**90** **100**	111 112	137 138	127 128	130 131	131 132	10.7 11.9	145 146	147 149	148 150	15.7 18.1
160	**100** **110**	168 169	205 206	198 200	200 202	209 210	12.6 13.9	230 234	233 236	241 244	18.8 21.4
180	**110** **125**	236 239	283 286	270 272	269 271	278 281	14.7 16.8	314 319	312 318	322 327	22.1 26.5
200	**125** **140**	306 309	361 364	348 351	346 349	358 361	19.0 21.5	369 376	367 373	380 386	28.6 33.5
220	**140** **160**	452	556	515	479	509	27.1 30.9	598	562	593	39.1 46.7
250	**160** **180**	582	710	664	618	649	32.7 36.9	784	739	770	48.5 56.9
280	**180** **200**	753	950	846	784	822	44.2 48.8	981	919	957	64.2 73.4
320	**200** **220**	1125	1404	1290	1180	1222	55.2 60.4	1452	1343	1385	79.8 90.2

[1] Weight without position measuring system [2] Weight with position measuring system

Areas, forces, flows

Piston	Piston rod	Area ratio	Areas			Force at 250 bar [1]			Flow at 0.1 m/s [2]		
			Piston	Rod	Annulus	Pressure	Diff.	Pulling	Out	Diff.	In
AL Ø mm	**MM** Ø mm	φ A_1/A_3	A_1 cm²	A_2 cm²	A_3 cm²	F_1 kN	F_2 kN	F_3 kN	q_{V1} L/min	q_{V2} L/min	q_{V3} L/min
40	**22**	1.43	12.56	3.80	8.76	31.40	9.50	21.90	7.5	2.3	5.3
	28	1.96		6.16	6.40		15.40	16.00		3.7	3.8
50	**28**	1.46	19.63	6.16	13.47	49.10	15.40	33.70	11.8	3.7	8.1
	36	2.08		10.18	9.45		25.45	23.65		6.1	5.7
63	**36**	1.48	31.17	10.18	20.99	77.90	25.45	52.45	18.7	6.1	12.6
	45	2.04		15.90	15.27		39.75	38.15		9.5	9.2
80	**45**	1.46	50.26	15.90	34.36	125.65	39.75	85.90	30.2	9.5	20.7
	56	1.96		24.63	25.63		61.55	64.10		14.8	15.4
100	**56**	1.46	78.54	24.63	53.91	196.35	61.55	134.80	47.1	14.8	32.3
	70	1.96		38.48	40.06		96.20	100.15		23.1	24.0
125	**70**	1.46	122.72	38.48	84.24	306.75	96.20	210.55	73.6	23.1	50.5
	90	2.08		63.62	59.10		159.05	147.70		38.2	35.4
140	**90**	1.70	153.94	63.62	90.32	384.75	159.05	225.70	92.4	38.2	54.2
	100	2.04		78.54	75.40		196.35	188.40		47.1	45.3
160	**100**	1.64	201.06	78.54	122.50	502.50	196.35	306.15	120.6	47.1	73.5
	110	1.90		95.06	106.00		237.65	264.85		57.0	63.6
180	**110**	1.60	254.47	95.06	159.43	636.17	237.65	398.52	152.7	57.0	95.7
	125	1.93		122.72	131.75		306.80	329.37		73.6	79.1
200	**125**	1.64	314.16	122.72	191.44	785.25	306.80	478.45	188.5	73.6	114.9
	140	1.96		153.96	160.20		384.90	400.35		92.4	96.1
220	**140**	1.68	380.1	153.9	226.2	950.3	384.8	565.5	228.1	92.4	135.7
	160	2.12		201.0	179.1		502.6	447.7		120.7	107.4
250	**160**	1.69	490.8	201.0	289.8	1227.2	502.7	724.5	294.5	120.7	173.8
	180	2.08		254.4	236.4		636.2	591.0		152.7	141.8
280	**180**	1.70	615.7	254.4	361.3	1539.4	636.2	903.2	369.4	152.7	216.7
	200	2.04		314.1	301.6		785.4	753.9		188.5	180.9
320	**200**	1.64	804.2	314.1	490.1	2010.6	785.4	1225.2	482.5	188.5	294.0
	220	1.90		380.1	424.2		950.3	1060.3		228.1	254.4





[1] Theoretical force
(efficiency not taken into account)

[2] Stroke velocity

Tolerances to ISO 8135

Installation dim.	WC	XC[2]	XO[2]	XS[1, 2]	XV[2]	ZP[2]	Stroke tolerance
Mounting style	MF3	MP3	MP5	MS2	MT4	MF4	
Stroke length	Tolerances						
≤ 1250	± 2	± 1.5	± 1,5	± 2	± 2	± 1.5	+ 2
> 1250 – ≤ 3150	± 4	± 3	± 3	± 4	± 4	± 3	+ 5
> 3150 – ≤ 8000	± 8	± 5	± 5	± 8	± 8	± 5	+ 8

[1] Not standardised [2] Stroke length included

IHC-Designer: Engineering software

The IHC-Designer (Interactive Hydraulics Cylinder Designer) is an engineering and selection aid for hydraulic cylinders. With the aid of the IHC-Designer it is possible for designers of machines and systems, by the logically guided type code questionnaire, to quickly and reliable find the optimum hydraulic cylinder solution. The software enables the design and engineering tasks to be carried out quicker and efficiently. After being guided through the product selection, the user quickly obtains reliable and exact technical data of the selected components as well as the associated 2D and 3D-CAD data in the correct file format for all current CAD systems.

You, as a user thereby reduce your costs and therefore increase your competitiveness.

Mounting style overview

CDH1 MP3
See pages 6, 7



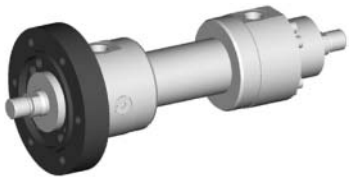
CDH1 MP5
See pages 8, 9



CDH1 MF3
See pages 10, 11



CGH1 MF3
See pages 10, 11



CDH1 MF4
See pages 12, 13



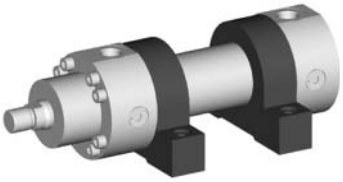
CDH1 MT4
See pages 14, 15



CGH1 MT4
See pages 14, 15



CDH1 MS2
See pages 16, 17



CGH1 MS2
See pages 16, 17



Ordering details

[1] = Only piston Ø 40 to 200 mm

[2] = The trunnion can be located as required.
Dim. „XV“ must always be stated, in case of an order, in clear text and in mm.

[3] = Only piston rod Ø 22 to 110 mm

[4] = Only piston rod Ø 22 to 140 mm

[5] = Not possible for MF4.

[6] = Standard for seal versions M, T, S and piston Ø 220 to 320 mm;
Not possible for seal versions A, B

[7] = For the CG version only one self-aligning clevis/plain clevis is fitted

[8] = Only MF3; MT4; MS2

[9] = Only possible in conjuction with position measuring sysem „T“

[10] = Only piston Ø 80 to 320 mm

[11] = Not for piston Ø 320 mm

[12] = For the CG version only on one piston rod end

[13] = Seal versions A, B is not possible;
Piston rod version „H“ is not possible;
Piston rod Ø 22 mm is not possible;
End position damping possible from piston rod Ø 45 mm;
CG version is not possible
Max. stroke length to be taken into account, page 21

[14] = Perm. stroke length to be taken into account, pages 31 to 33

[15] = Not possible for version piston rod end N

Ordering details

H1 / / / / / A / / / / / / / / /

Differential cylinder = CD
Double rod cylinder [8] = CG

Series = H1

Mounting styles
Plain clevis at base [1] = MP3
Self-aligning clevis at base = MP5
Round flange at head = MF3
Round flange at base = MF4
Trunnions [2] = MT4
Foot mounting = MS2

Piston Ø (40 to 320 mm)
See page 2

Piston rod Ø (22 to 220 mm)
See page 2

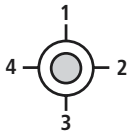
Stroke length in mm [14]

Design principle
Head and base flanged = A

Component series
10 to 19 unchanged installation and connection dimensions = 1X
Only piston Ø 40 to 200 mm
20 to 29 unchanged installation and connection dimensions = 2X
Only piston Ø 220 to 320 mm

Connection ports/versions
Pipe thread to ISO 228/1 = B
Metric ISO thread = M
Flange hole pattern to ISO 6162 Tab.2 (≙SAE 6000 PSI) [5], [10] = D
Flange hole pattern to ISO 6164 Tab.2 [5] = H
Pipe thread to ISO 228/1 with a ring flange that has a machined flat = C

Connection port/position at head and base = 1 / = 2 / = 3 / = 4



Viewed on the piston rod

Piston rod version
Hard chromium plated = C
Hardened and hard chromium plated [3] = H
Nickel plated and hard chromium plated [4] = N

Option 2
A = [15] Maintenance-free plain bearing
B = Flange grease nipple
C = [9] Analogue output 4-20 mA
F = [9] Analogue output 0-10 V
D = [9] Digital output SSI
Y = [12] Enter piston rod extension LY in clear text in mm
W = Without option

Option 1
A = Screwed coupling, on both sides
F = [6] Guide rings
E = Inductive proximity switch **without** plug-in connector
Plug-in connector – separate order, see page 24
T = [13] Pos. measuring system (magnetostrictive) **without** plug-in connector
Plug-in connector – separate order, see page 23
W = Without option

Seal version
Suitable for mineral oil to DIN 51524 HL, HLP and HFA
M = Standard seal system
T = Servo quality/reduced friction
A = Chevron seal kits
Suitable for phosphate ester HFD-R
S = Servo quality/reduced friction
B = Chevron seal kits

End position cushioning
U = Without
D [1] = Both ends, self adjusting
E = Both ends, adjustable

Piston rod end
A = Thread for self-aligning clevis CGAS
G = [11] Thread for self-aligning clevis CGA, CGAK, plain clevis CSA
S = [7] With mounted self-aligning clevis CGAS
L = [7], [11] With mounted self-aligning clevis CGA
M = [7], [11] With mounted self-aligning clevis CGAK
N = [1], [7] With mounted plain clevis CSA

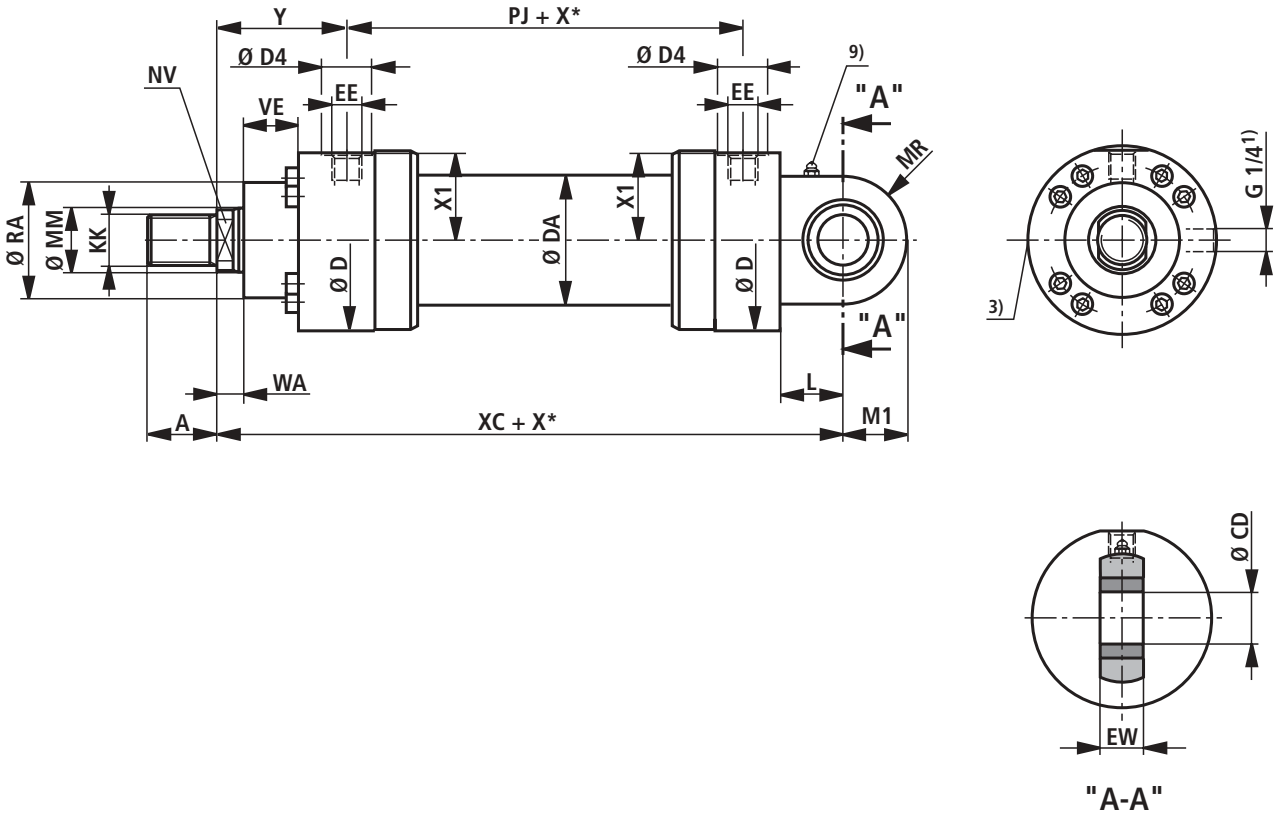
Ordering example:

CDH1 MT4/63/36/350A1X/B1CGDMWW, XV = 300 mm

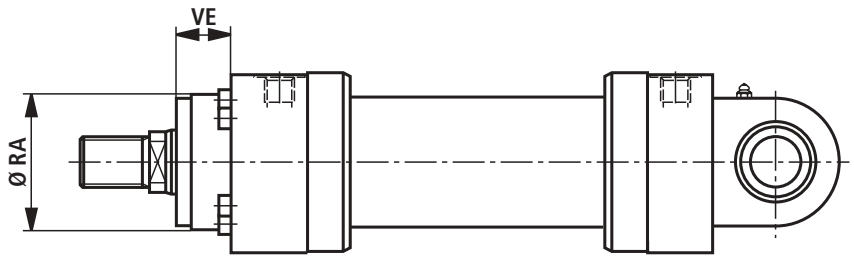
CGH1 MF3/100/56/500A1X/B1CGUMWW

Plain clevis at base MP3

CDH1 MP3; AL-Ø 40-200 mm



For seal versions „A", „B" and AL-Ø 160 - 200 mm



Dimensions MP3 (nominal dimensions in mm)

AL Ø	MM Ø	KK 5)	A 5)	KK 6)	A 6)	NV	D	DA	D4 2)	EE 4)	EE 4)	Y	PJ
40	**22/28**	M16x1.5	16	M18x2	30	16/22	88	50	34	G 1/2	M22x1.5	79	120
50	**28/36**	M22x1.5	22	M24x2	35	22/30	102	60	34	G 1/2	M22x1.5	87	120
63	**36/45**	M28x1.5	28	M30x2	45	30/36	120	78	42	G 3/4	M27x2	100	133
80	**45/56**	M35x1.5	35	M39x3	55	36/46	140	95	42	G 3/4	M27x2	104	146
100	**56/70**	M45x1.5	45	M50x3	75	46/60	170	125	47	G 1	M33x2	124	171
125	**70/90**	M58x1.5	58	M64x3	95	60/75	206	150	58	G 1 1/4	M42x2	135	205
140	**90/100**	M65x1.5	65	M80x3	110	75/85	226	170	58	G 1 1/4	M42x2	156	219
160	**100/110**	M80x2	80	M90x3	120	85/95	265	190	65	G 1 1/2	M48x2	185	240
180	**110/125**	M100x2	100	M100x3	140	95/110	292	210	65	G 1 1/2	M48x2	199	264
200	**125/140**	M110x2	110	M110x4	150	110/120	310	235	65	G 1 1/2	M48x2	205	278

AL Ø	MM Ø	X1	WA	XC	L	MR	M1	CD H11	EW −0.4	RA[7] f8	VE[7]	RA[8] f8	VE[8]
40	**22/28**	41	14	252	32.5	31	28	25	23	52	40	52	20
50	**28/36**	48.5	18	265	37.5	36	32.5	30	28	65	40	65	16
63	**36/45**	56.5	22	302	45	42	40	35	30	75	45	75	17
80	**45/56**	67	20	330	50	52	50	40	35	95	45	95	13
100	**56/70**	82	30	385	60	65	62.5	50	40	115	55	115	20
125	**70/90**	99	32	447	70	70	70	60	50	135	60	135	17
140	**90/100**	109.5	35	490	75	82	82	70	55	155	70	155	22
160	**100/110**	129	40	550	85	95	95	80	60	200	80	200	80
180	**110/125**	142.5	40	610	90	113	113	90	65	220	90	220	90
200	**125/140**	152	40	645	115	125	125	100	70	235	95	235	95

AL = Piston Ø

MM = Piston rod Ø

X* = Stroke length

1) = Bleeding: When viewed on the piston rod, the orientation is always offset by 90° to the pipe connection (in a clockwise direction)

2) = Ø D4 max. 0.5 mm deep

3) = Throttle valve only with end position damping „E” (180° to the bleed point)

4) = For flange connections see separate table on pages 18 and 19

5) = Thread version „G”

6) = Thread version „A”

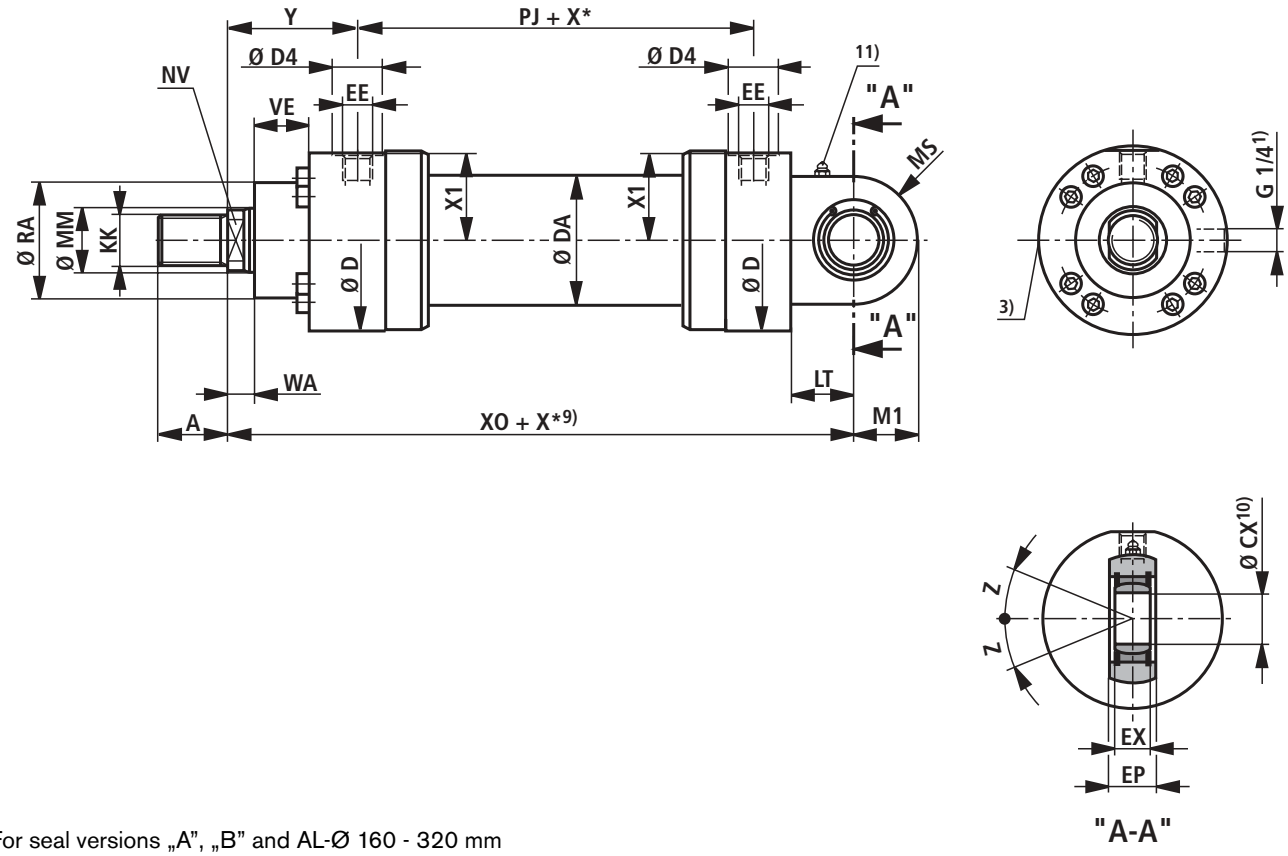
7) = Dimension for cylinder with seal versions M, T and S

8) = Dimension for cylinder with seal versions A and B

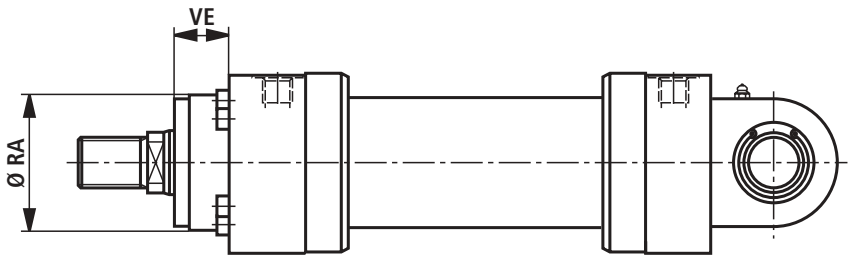
9) = Grease nipple; cone head form A to DIN 71412

Self-aligning clevis at base MP5

CDH1 MP5



For seal versions „A", „B" and AL-Ø 160 - 320 mm



Dimensions MP5 (nominal dimensions in mm)

AL Ø	MM Ø	KK 5)	A 5)	KK 6)	A 6)	NV	D	DA	D4 2)	EE 4)	EE 4)	Y	PJ	X1
40	**22/28**	M16x1.5	16	M18x2	30	16/22	88	50	34	G 1/2	M22x1.5	79	120	41
50	**28/36**	M22x1.5	22	M24x2	35	22/30	102	60	34	G 1/2	M22x1.5	87	120	48.5
63	**36/45**	M28x1.5	28	M30x2	45	30/36	120	78	42	G 3/4	M27x2	100	133	56.5
80	**45/56**	M35x1.5	35	M39x3	55	36/46	140	95	42	G 3/4	M27x2	104	146	67
100	**56/70**	M45x1.5	45	M50x3	75	46/60	170	125	47	G 1	M33x2	124	171	82
125	**70/90**	M58x1.5	58	M64x3	95	60/75	206	150	58	G 1 1/4	M42x2	135	205	99
140	**90/100**	M65x1.5	65	M80x3	110	75/85	226	170	58	G 1 1/4	M42x2	156	219	109.5
160	**100/110**	M80x2	80	M90x3	120	85/95	265	190	65	G 1 1/2	M48x2	185	240	129
180	**110/125**	M100x2	100	M100x3	140	95/110	292	210	65	G 1 1/2	M48x2	199	264	142.5
200	**125/140**	M110x2	110	M110x4	150	110/120	310	235	65	G 1 1/2	M48x2	205	278	152
220	**140/160**	M120x3	120	M120x4	160	120/140	355	273	65	G 1 1/2	M48x2	242	326	174
250	**160/180**	M120x3	120	M120x4	160	140/160	393	305	65	G 1 1/2	M48x2	266	326	194
280	**180/200**	M130x3	130	M150x4	190	160/180	425	343	65	G 1 1/2	M48x2	282	375	210
320	**200/220**	–	–	M160x4	200	180/200	490	394	65	G 1 1/2	M48x2	287	431	242

AL Ø	MM Ø	WA	XO	X* min	LT	M1	MS	CX	EP – 0.4	EX	Z	RA 7) f8	VE 7)	RA 8) f8	VE 8)
40	**22/28**	14	252	–	32.5	28	31	$25_{-0.010}$	23	$20_{-0.12}$	7°	52	40	52	20
50	**28/36**	18	265	–	37.5	32.5	36	$30_{-0.010}$	28	$22_{-0.12}$	6°	65	40	65	16
63	**36/45**	22	302	–	45	40	42	$35_{-0.012}$	30	$25_{-0.12}$	6°	75	45	75	17
80	**45/56**	20	330	–	50	50	52	$40_{-0.012}$	35	$28_{-0.12}$	7°	95	45	95	13
100	**56/70**	30	385	–	60	62.5	65	$50_{-0.012}$	40	$35_{-0.12}$	6°	115	55	115	20
125	**70/90**	32	447	–	70	70	70	$60_{-0.015}$	50	$44_{-0.15}$	6°	135	60	135	17
140	**90/100**	35	490	–	75	82	82	$70_{-0.015}$	55	$49_{-0.15}$	6°	155	70	155	22
160	**100/110**	40	550	–	85	95	95	$80_{-0.015}$	60	$55_{-0.15}$	6°	200	80	200	80
180	**110/125**	40	610	–	90	113	113	$90_{-0.020}$	65	$60_{-0.20}$	5°	220	90	220	90
200	**125/140**	40	645	–	115	125	125	$100_{-0.020}$	70	$70_{-0.20}$	7°	235	95	235	95
220	**140/160**	40	750	–	125	142.5	132,5	$110_{-0.020}$	80	$70_{-0.20}$	6°	270	115	270	115
250	**160/180**	40	789	–	140	160	150	$110_{-0.020}$	80	$70_{-0.20}$	6°	300	125	300	125
280	**180/200**	40	884	31	150	180	170	$120_{-0.020}$	90	$85_{-0.20}$	6°	325	130	325	130
320	**200/220**	40	980	–	175	200	190	$140_{-0.020}$	110	$90_{-0.20}$	7°	365	155	365	155

AL = Piston Ø

MM = Piston rod Ø

X* = Stroke length

1) = Bleeding: When viewed on the piston rod, the orientation is always offset by 90° to the pipe connection (in a clockwise direction)

2) = Ø D4 max. 0,5 mm deep

3) = Throttle valve only with end position damping „E" (180° to the bleed point)

4) = For flange connections see separate table on pages 18 and 19

5) = Thread version „G"

6) = Thread version „A"

7) = Dimensions for cylinder with seal versions M, T and S

8) = Dimensions for cylinder with seal versions A and B

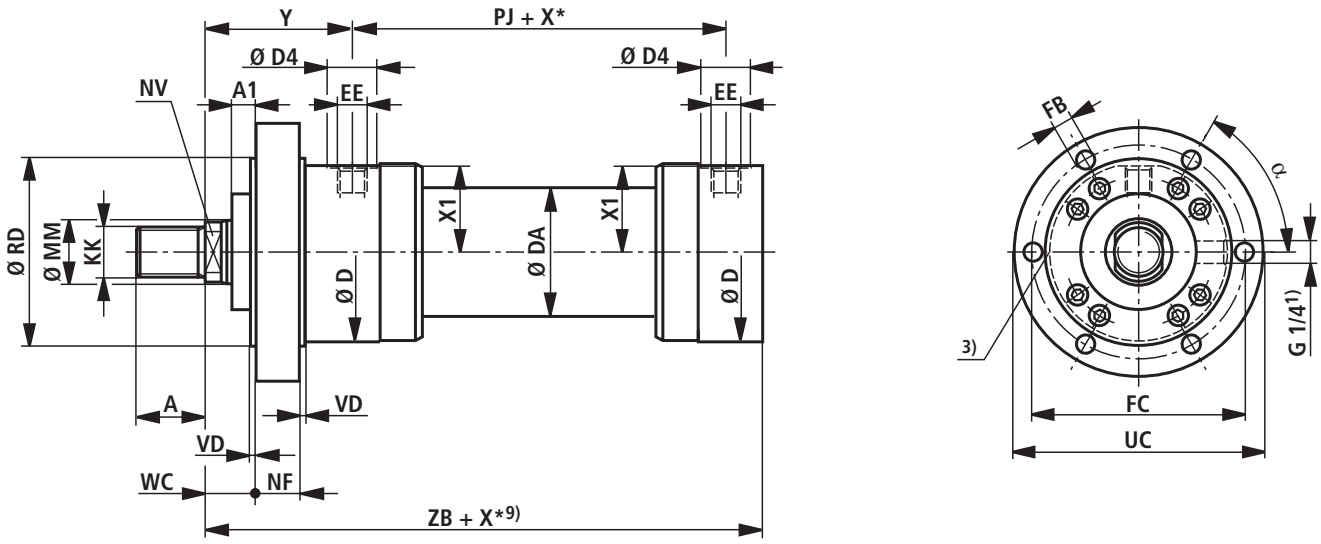
9) = Take the minimum stroke length „X*min." into account

10) = Associated pin Ø m6; associated pin Ø j6 with maintenance-free self-aligning clevis

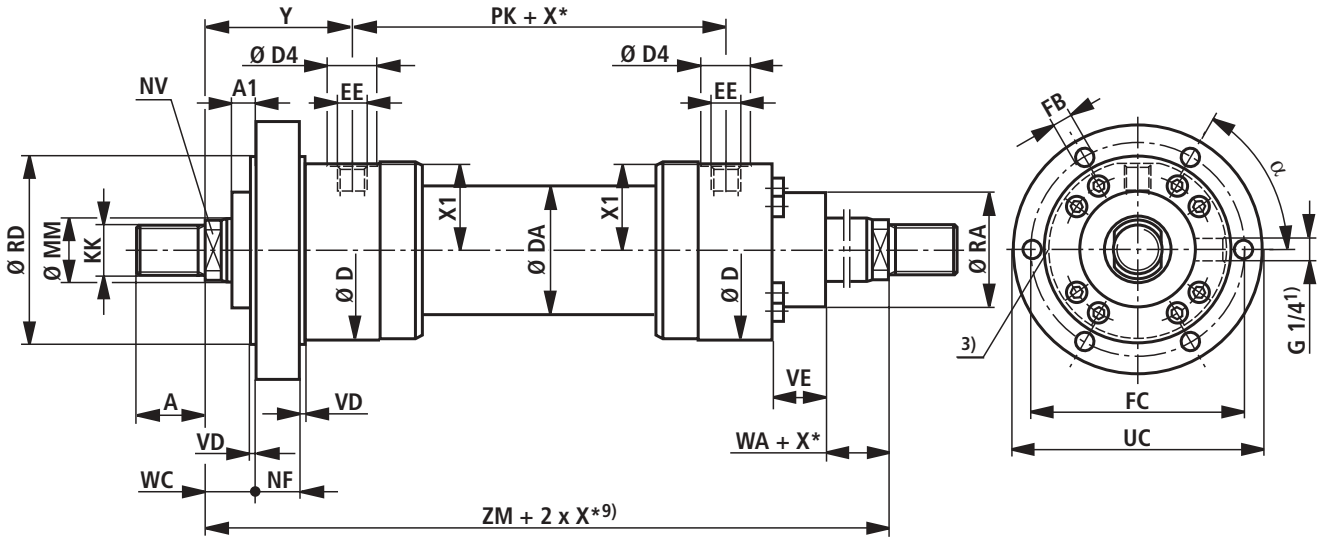
11) = Grease nipple; cone head form A to DIN 71412

Round flange at head MF3

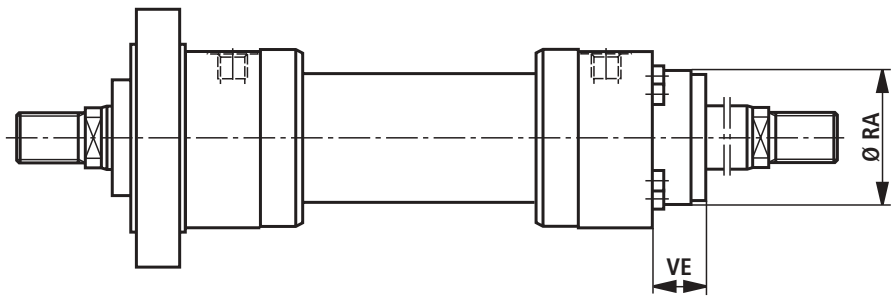
CDH1 MF3



CGH1 MF3



For seal versions „A", „B" and AL-Ø 160 - 320 mm



Dimensions MF3 (nominal dimensions in mm)

AL Ø	MM Ø	KK 5)	A 5)	KK 6)	A 6)	NV	D	DA	D4 2)	EE 4)	EE 4)	Y	PJ	X1
40	**22/28**	M16x1.5	16	M18x2	30	16/22	88	50	34	G 1/2	M22x1.5	79	120	41
50	**28/36**	M22x1.5	22	M24x2	35	22/30	102	60	34	G 1/2	M22x1.5	87	120	48.5
63	**36/45**	M28x1.5	28	M30x2	45	30/36	120	78	42	G 3/4	M27x2	100	133	56.5
80	**45/56**	M35x1.5	35	M39x3	55	36/46	140	95	42	G 3/4	M27x2	104	146	67
100	**56/70**	M45x1.5	45	M50x3	75	46/60	170	125	47	G 1	M33x2	124	171	82
125	**70/90**	M58x1.5	58	M64x3	95	60/75	206	150	58	G 1 1/4	M42x2	135	205	99
140	**90/100**	M65x1.5	65	M80x3	110	75/85	226	170	58	G 1 1/4	M42x2	156	219	109.5
160	**100/110**	M80x2	80	M90x3	120	85/95	265	190	65	G 1 1/2	M48x2	185	240	129
180	**110/125**	M100x2	100	M100x3	140	95/110	292	210	65	G 1 1/2	M48x2	199	264	142.5
200	**125/140**	M110x2	110	M110x4	150	110/120	310	235	65	G 1 1/2	M48x2	205	278	152
220	**140/160**	M120x3	120	M120x4	160	120/140	355	273	65	G 1 1/2	M48x2	242	326	174
250	**160/180**	M120x3	120	M120x4	160	140/160	395	305	65	G 1 1/2	M48x2	266	326	194
280	**180/200**	M130x3	130	M150x4	190	160/180	425	343	65	G 1 1/2	M48x2	282	375	210
320	**200/220**	–	–	M160x4	200	180/200	490	394	65	G 1 1/2	M48x2	287	431	243

AL Ø	MM Ø	RD e8	WC	VD	NF	PK	A1	ZB	ZM	X* min	FB H13	FC js13	UC −1	α	WA	RA 7) f8	VE 7)	RA 8) f8	VE 8)
40	**22/28**	90	19	5	30	120	0	226	278	–	9	108	130	60°	14	52	40	52	20
50	**28/36**	110	23	5	30	120	0	233	294	–	11	130	160	60°	18	65	40	65	16
63	**36/45**	130	27	5	35	133	0	262	333	–	13.5	155	185	60°	22	75	45	75	17
80	**45/56**	145	25	5	35	146	0	280	354	–	13.5	170	200	60°	20	95	45	95	13
100	**56/70**	175	35	5	45	171	0	330	419	–	17.5	205	245	60°	30	115	55	115	20
125	**70/90**	210	37	5	50	205	0	382	475	–	22	245	295	60°	32	135	60	135	17
140	**90/100**	230	45	10	50	219	0	420	531	–	22	265	315	60°	35	155	70	155	22
160	**100/110**	275	50	10	60	240	0	475	610	–	30	325	385	60°	40	200	80	200	80
180	**110/125**	300	50	10	70	264	0	515	662	–	30	360	420	60°	40	220	90	220	90
200	**125/140**	320	50	10	75	278	0	535	688	–	33	375	445	60°	40	235	95	235	95
220	**140/160**	370	60	10	85	326	20	635	810	–	33	430	490	60°	40	270	115	270	115
250	**160/180**	415	70	10	85	326	30	659	858	–	39	485	555	60°	40	300	125	300	125
280	**180/200**	450	65	10	95	375	25	744	939	31	39	520	590	60°	40	325	130	325	130
320	**200/220**	510	65	10	120	431	25	815	1005	–	45	600	680	60°	40	365	155	365	155

AL = Piston Ø

MM = Piston rod Ø

X* = Stroke length

1) = Bleeding: When viewed on the piston rod, the orientation is always offset by 90° to the pipe connection (in a clockwise direction)

2) = Ø D4 max. 0.5 mm deep

3) = Throttle valve only with end position damping „E" (180° to the bleed position)

4) = For flange connections see separate table on pages 18 and 19

5) = Thread version G"

6) = Thread version „A"

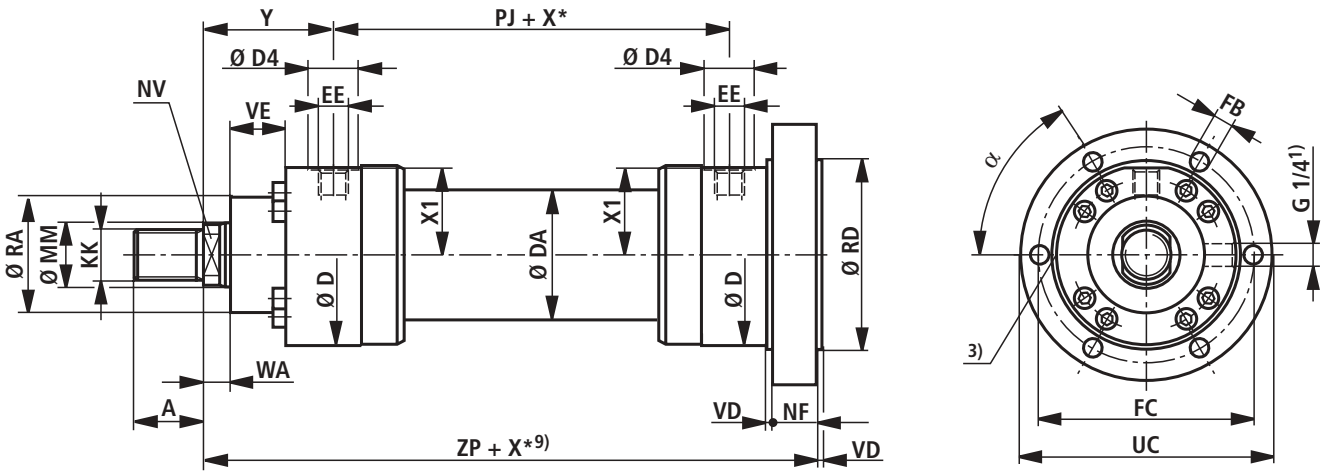
7) = Dimensions for cylinder with seal versions M, T and S

8) = Dimensions for cylinder with seal versions A and B

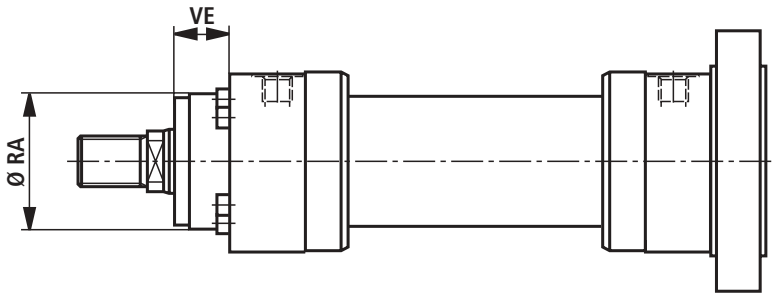
9) = Take the minimum stroke length „X*min." into account

Round flange at base MF4

CDH1 MF4



For seal versions „A", „B" and AL-Ø 160 - 320 mm



Dimensions MF4 (nominal dimensions in mm)

AL Ø	MM Ø	KK [5]	A [5]	KK [6]	A [6]	NV	D	DA	D4 [2]	EE [4]	EE [4]	Y	PJ	X1
40	**22/28**	M16x1.5	16	M18x2	30	16/22	88	50	34	G 1/2	M22x1.5	79	120	41
50	**28/36**	M22x1.5	22	M24x2	35	22/30	102	60	34	G 1/2	M22x1.5	87	120	48.5
63	**36/45**	M28x1.5	28	M30x2	45	30/36	120	78	42	G 3/4	M27x2	100	133	56.5
80	**45/56**	M35x1.5	35	M39x3	55	36/46	140	95	42	G 3/4	M27x2	104	146	67
100	**56/70**	M45x1.5	45	M50x3	75	46/60	170	125	47	G 1	M33x2	124	171	82
125	**70/90**	M58x1.5	58	M64x3	95	60/75	206	150	58	G 1 1/4	M42x2	135	205	99
140	**90/100**	M65x1.5	65	M80x3	110	75/85	226	170	58	G 1 1/4	M42x2	156	219	109.5
160	**100/110**	M80x2	80	M90x3	120	85/95	265	190	65	G 1 1/2	M48x2	185	240	129
180	**110/125**	M100x2	100	M100x3	140	95/110	292	210	65	G 1 1/2	M48x2	199	264	142.5
200	**125/140**	M110x2	110	M110x4	150	110/120	310	235	65	G 1 1/2	M48x2	205	278	152
220	**140/160**	M120x3	120	M120x4	160	120/140	355	273	65	G 1 1/2	M48x2	242	326	174
250	**160/180**	M120x3	120	M120x4	160	140/160	393	305	65	G 1 1/2	M48x2	266	326	194
280	**180/200**	M130x3	130	M150x4	190	160/180	425	343	65	G 1 1/2	M48x2	282	375	210
320	**200/220**	–	–	M160x4	200	180/200	490	394	65	G 1 1/2	M48x2	287	431	243

AL Ø	MM Ø	WA	ZP	X* min	NF	VD	RD e8	FB H13	FC js13	UC –1	α	RA [7] f8	VE [7]	RA [8] f8	VE [8]
40	**22/28**	14	256	–	30	5	90	9	108	130	60°	52	40	52	20
50	**28/36**	18	264	–	30	5	110	11	130	160	60°	65	40	65	16
63	**36/45**	22	297	–	35	5	130	13.5	155	185	60°	75	45	75	17
80	**45/56**	20	315	–	35	5	145	13.5	170	200	60°	95	45	95	13
100	**56/70**	30	375	–	45	5	175	17.5	205	245	60°	115	55	115	20
125	**70/90**	32	432	–	50	5	210	22	245	295	60°	135	60	135	17
140	**90/100**	35	475	–	50	10	230	22	265	315	60°	155	70	155	22
160	**100/110**	40	535	–	60	10	275	30	325	385	60°	200	80	200	80
180	**110/125**	40	585	–	70	10	300	30	360	420	60°	220	90	220	90
200	**125/140**	40	615	–	75	10	320	33	375	445	60°	235	95	235	95
220	**140/160**	40	720	–	85	10	370	33	430	490	60°	270	115	270	115
250	**160/180**	40	744	–	85	10	415	39	485	555	60°	300	125	300	125
280	**180/200**	40	839	31	95	10	450	39	520	590	60°	325	130	325	130
320	**200/220**	40	935	–	120	10	510	45	600	680	60°	365	155	365	155

AL = Piston Ø

MM = Piston rod Ø

X* = Stroke length

[1] = Bleeding: When viewed on the piston rod, the orientation is always offset by 90° to the pipe connection (in a clockwise direction)

[2] = Ø D4 max. 0.5 mm deep

[3] = Throttle valve only with end position damping „E" (180° to the bleed position)

[4] = For flange connections see separate table on pages 18 and 19

[5] = Thread version „G"

[6] = Thread version „A"

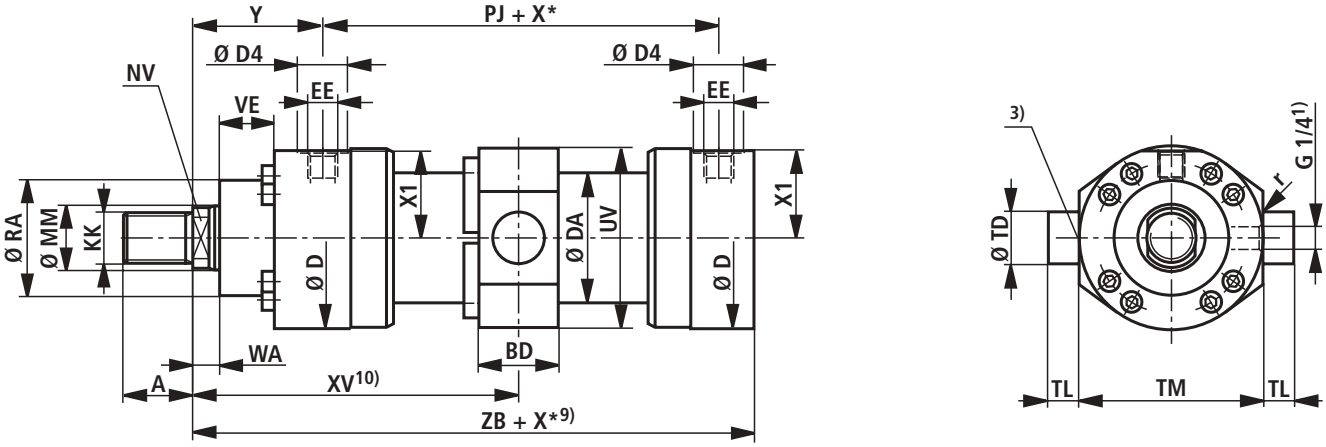
[7] = Dimensions for cylinder with seal versions M, T and S

[8] = Dimensions for cylinder with seal versions A and B

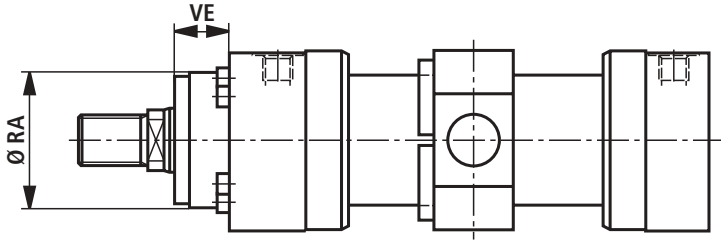
[9] = Take the minimum stroke length „X*min." into account

Trunnions MT4

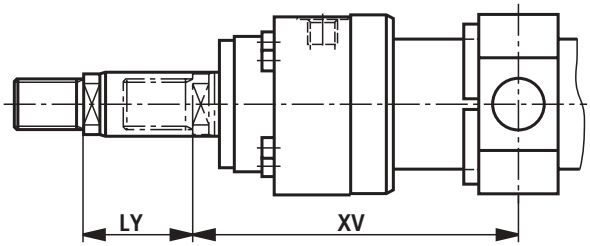
CDH1 MT4



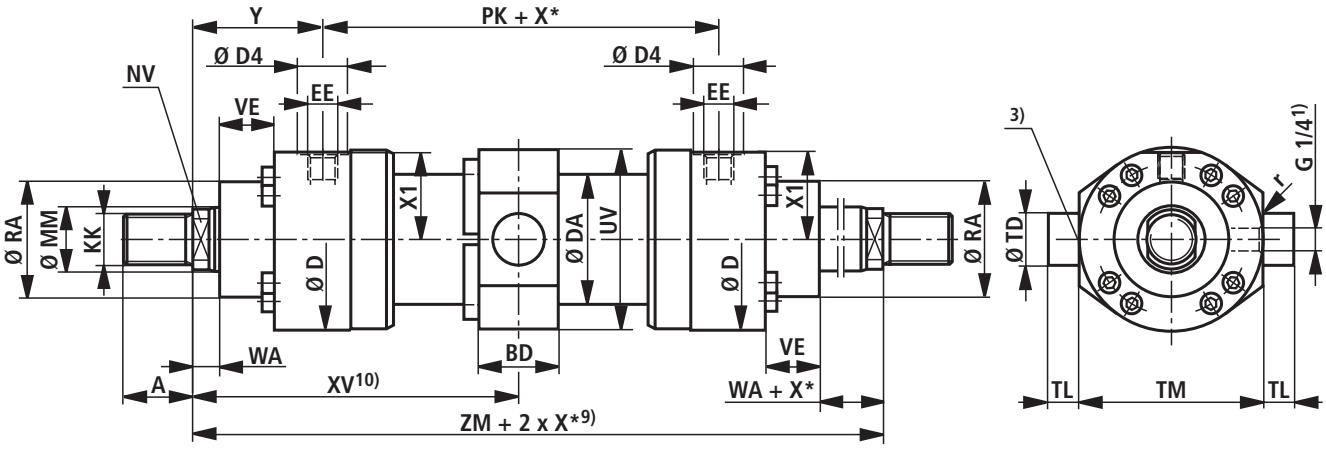
For seal versions „A", „B" and AL-Ø 160 - 320 mm



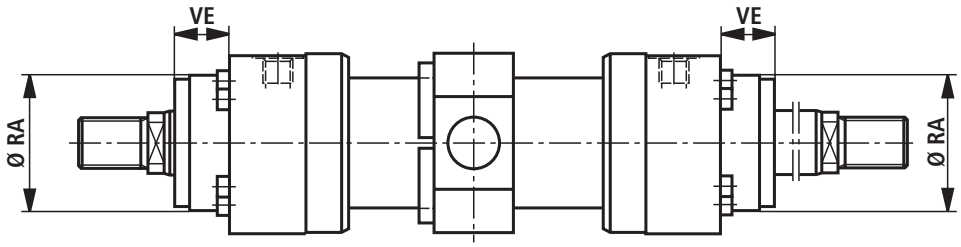
Dimensions for a cylinder with a rod extension „LY" in the retracted condition



CGH1 MT4



For seal versions „A", „B" and AL-Ø 160 - 320 mm



Dimensions MT4 (nominal dimensions in mm)

AL Ø	MM Ø	KK 5)	A 5)	KK 6)	A 6)	NV	D	DA	D4 2)	EE 4)	EE 4)	Y	PJ	X1	WA
40	**22/28**	M16x1.5	16	M18x2	30	16/22	88	50	34	G 1/2	M22x1.5	79	120	41	14
50	**28/36**	M22x1.5	22	M24x2	35	22/30	102	60	34	G 1/2	M22x1.5	87	120	48.5	18
63	**36/45**	M28x1.5	28	M30x2	45	30/36	120	78	42	G 3/4	M27x2	100	133	56.5	22
80	**45/56**	M35x1.5	35	M39x3	55	36/46	140	95	42	G 3/4	M27x2	104	146	67	20
100	**56/70**	M45x1.5	45	M50x3	75	46/60	170	125	47	G 1	M33x2	124	171	82	30
125	**70/90**	M58x1.5	58	M64x3	95	60/75	206	150	58	G1 1/4	M42x2	135	205	99	32
140	**90/100**	M65x1.5	65	M80x3	110	75/85	226	170	58	G1 1/4	M42x2	156	219	109.5	35
160	**100/110**	M80x2	80	M90x3	120	85/95	265	190	65	G1 1/2	M48x2	185	240	129	40
180	**110/125**	M100x2	100	M100x3	140	95/110	292	210	65	G1 1/2	M48x2	199	264	142.5	40
200	**125/140**	M110x2	110	M110x4	150	110/120	310	235	65	G1 1/2	M48x2	205	278	152	40
220	**140/160**	M120x3	120	M120x4	160	120/140	355	273	65	G1 1/2	M48x2	242	326	174	40
250	**160/180**	M120x3	120	M120x4	160	140/160	395	305	65	G1 1/2	M48x2	266	326	194	40
280	**180/200**	M130x3	130	M150x4	190	160/180	425	343	65	G1 1/2	M48x2	282	375	210	40
320	**200/220**	–	–	M160x4	200	180/200	490	394	65	G1 1/2	M48x2	287	431	243	40

AL Ø	MM Ø	PK	ZB	ZM	X* min.	XV[11] mitt	XV[10] min.	XV[10] max.	BD	UV	TD e8	TL js16	TM h13	r	RA[7] f8	VE[7]	RA[8] f8	VE[8]
40	**22/28**	120	226	278	22	139+X*/2	150	136+X*	38	88	30	20	95	1.6	52	40	52	20
50	**28/36**	120	233	294	32	147+X*/2	163	140+X*	38	102	30	20	115	1.6	65	40	65	16
63	**36/45**	133	262	333	47	166.5+X*/2	190	155+X*	48	120	35	20	130	2	75	45	75	17
80	**45/56**	146	280	354	58	177+X*/2	206	160+X*	58	140	40	25	145	2	95	45	95	13
100	**56/70**	171	330	419	79	209.5+X*/2	249	185+X*	78	170	50	30	175	2	115	55	115	20
125	**70/90**	205	382	475	91	237.5+X*/2	283	207+X*	98	206	60	40	210	2.5	135	60	135	17
140	**90/100**	219	420	531	121	265.5+X*/2	326	220+X*	118	226	65	42.5	230	2.5	155	70	155	22
160	**100/110**	240	475	610	142	305+X*/2	376	254+X*	128	265	75	52.5	275	2.5	200	80	200	80
180	**110/125**	264	515	661	158	331+X*/2	410	272+X*	138	292	85	55	300	2.5	220	90	220	90
200	**125/140**	278	535	688	194	344+X*/2	441	267+X*	168	310	90	55	320	2.5	235	95	235	95
220	**125/140**	326	635	810	155	405+X*/2	482.5	327.5+X*	135	355	100	60	370	2.5	270	115	270	115
250	**160/180**	326	659	858	175	429+X*/2	516.5	341.5+X*	145	395	110	65	410	2.5	300	125	300	125
280	**180/200**	375	744	939	336	469.5+X*/2	637.5	301.5+X*	165	425	130	70	450	2.5	325	130	325	130
320	**200/220**	431	815	1005	180	502.5+X*/2	592.5	412.5+X*	195	490	160	90	510	2.5	365	155	365	155

AL = Piston Ø

MM = Piston rod Ø

X* = Stroke length

1) = Bleeding: When viewed on the piston rod, the orientation is always offset by 90° to the pipe connection (in a clockwise direction)

2) = Ø D4 max. 0.5 mm deep

3) = Throttle valve only with end position damping „E" (180° to the bleed point)

4) = For flange connections see separate table on pages18 and 19

5) = Thread version „G"

6) = Thread version „A"

7) = Dimensions for cylinder with seal versions M, T and S

8) = Dimensions for cylinder with seal versions A and B

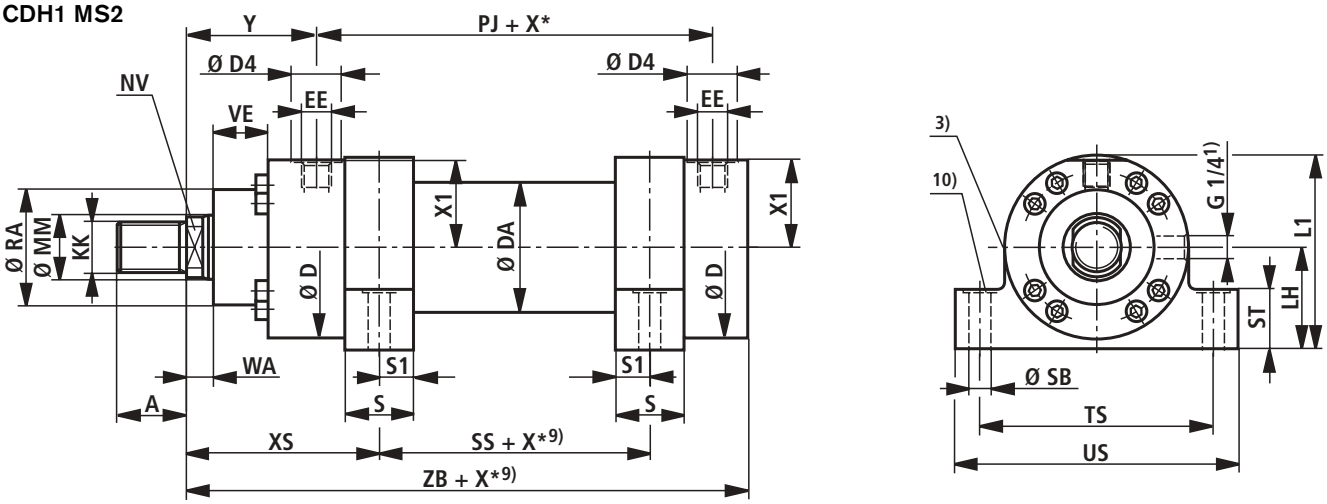
9) = Take the minimum stroke length „X*min." into account

10) = Dim. „XV" must always be stated in clear text in the case of an order.
Preferred XV dim.: The trunnions located in the middle of the cylinder. Take the XVmin and XVmax. into account.

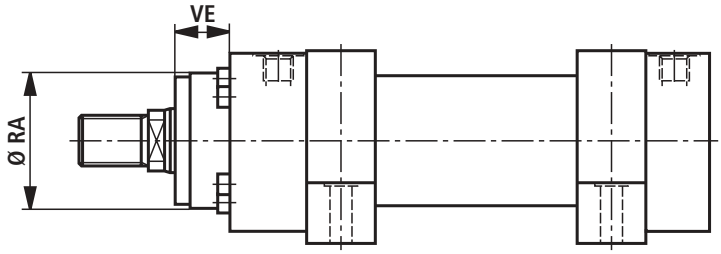
11) = XVmid. recommendation:
The trunnions located in the middle of the cylinder

Foot mounting MS2

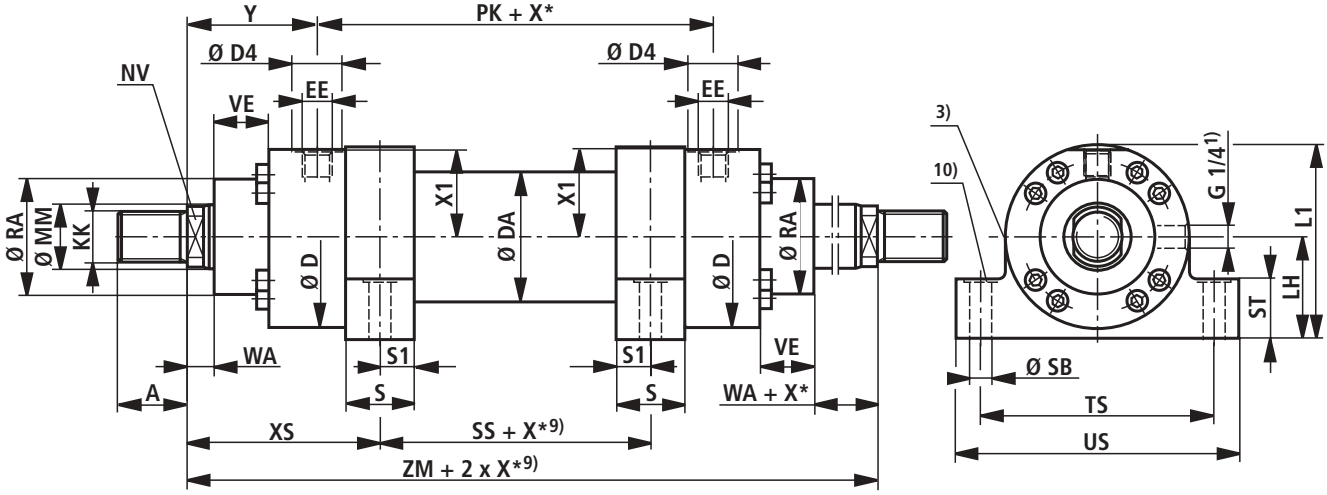
CDH1 MS2



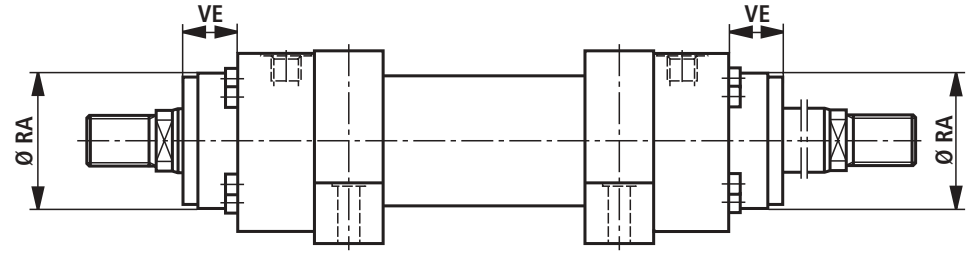
For seal versions „A", „B" and AL-Ø 160 - 320 mm



CGH1 MS2



For seal versions „A", „B" and AL-Ø 160 - 320 mm



Dimensions MS2 (nominal dimensions in mm)

AL Ø	MM Ø	KK 5)	A 5)	KK 6)	A 6)	NV	D	DA	D4 2)	EE 4)	EE 4)	Y	PJ	X1	WA
40	**22/28**	M16x1.5	16	M18x2	30	16/22	88	50	34	G 1/2	M22x1.5	79	120	41	14
50	**28/36**	M22x1.5	22	M24x2	35	22/30	102	60	34	G 1/2	M22x1.5	87	120	48.5	18
63	**36/45**	M28x1.5	28	M30x2	45	30/36	120	78	42	G 3/4	M27x2	100	133	56.5	22
80	**45/56**	M35x1.5	35	M39x3	55	36/46	140	95	42	G 3/4	M27x2	104	146	67	20
100	**56/70**	M45x1.5	45	M50x3	75	46/60	170	125	47	G 1	M33x2	124	171	82	30
125	**70/90**	M58x1.5	58	M64x3	95	60/75	206	150	58	G1 1/4	M42x2	135	205	99	32
140	**90/100**	M65x1.5	65	M80x3	110	75/85	226	170	58	G1 1/4	M42x2	156	219	109.5	35
160	**100/110**	M80x2	80	M90x3	120	85/95	265	190	65	G1 1/2	M48x2	185	240	129	40
180	**110/125**	M100x2	100	M100x3	140	95/110	292	210	65	G1 1/2	M48x2	199	264	142.5	40
200	**125/140**	M110x2	110	M110x4	150	110/120	310	235	65	G1 1/2	M48x2	205	278	152	40
220	**140/160**	M120x3	120	M120x4	160	120/140	355	273	65	G1 1/2	M48x2	242	326	174	40
250	**160/180**	M120x3	120	M120x4	160	140/160	395	305	65	G1 1/2	M48x2	266	326	194	40
280	**180/200**	M130x3	130	M150x4	190	160/180	425	343	65	G1 1/2	M48x2	282	375	210	40
320	**200/220**	–	–	M160x4	200	180/200	490	394	65	G1 1/2	M48x2	287	431	243	40

AL Ø	MM Ø	PK	XS	ZB	ZM	SS	X* min.	S	S1	SB H13	ST	TS js13	US −1	LH	L1	RA 7) f8	VE 7)	RA 8) f8	VE 8)
40	**22/28**	120	114	226	278	50	–	30	15	11	32	110	135	45	89	52	40	52	20
50	**28/36**	120	124.5	233	294	45	–	35	17.5	11	37	130	155	55	106	65	40	65	16
63	**36/45**	133	142	262	333	49	–	40	20	13.5	42	150	180	65	125	75	45	75	17
80	**45/56**	146	151	280	354	52	2	50	25	17.5	47	180	220	75	145	95	45	95	13
100	**56/70**	171	179	330	419	61	3	60	30	22	57	210	255	90	175	115	55	115	20
125	**70/90**	205	200	382	475	75	–	70	35	26	67	255	305	105	208	135	60	135	17
140	**90/100**	219	230.5	420	531	70	19	85	42.5	30	72	290	350	115	228	155	70	155	22
160	**100/110**	240	272.5	475	610	65	44	105	52.5	33	77	330	400	135	267.5	200	80	200	80
180	**110/125**	264	296.5	515	662	69	50	115	57.5	40	92	360	440	150	296	220	90	220	90
200	**125/140**	278	307.5	535	688	73	56	125	62.5	40	97	385	465	160	315	235	95	235	95
220	**140/160**	326	367.5	635	810	75	100	155	77.5	45	102	445	530	185	362.5	270	115	270	115
250	**160/180**	326	391.5	659	858	75	100	155	77.5	52	112	500	600	205	402.5	300	125	300	125
280	**180/200**	375	407.5	744	939	124	171	155	77.5	52	127	530	630	225	437.5	325	130	325	130
320	**200/220**	431	440	815	1005	125	85	190	95	62	142	610	730	255	500	365	155	365	155

AL = Piston Ø

MM = Piston rod Ø

X* = Stroke length

1) = Bleeding: When viewed on the piston rod the orientation is always offset by 90° to the pipe connection (in a clockwise direction)

2) = Ø D4 max. 0.5 mm deep

3) = Throttle valve only with end position damping „E" (180° to the bleed point)

4) = For flange connections see separate table on pages 18 and 19

5) = Thread version „G"

6) = Thread version „A"

7) = Dimensions for cylinder with seal versions M, T and S

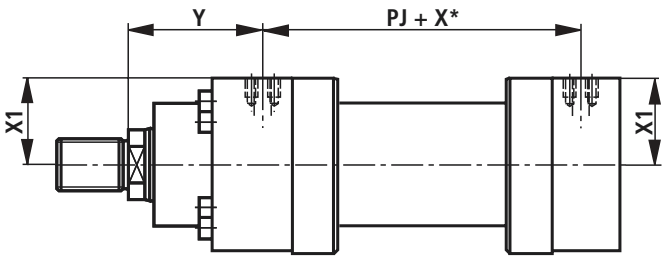
8) = Dimensions for cylinder with seal versions A and B

9) = Take the minimum stroke length „X*min." into account

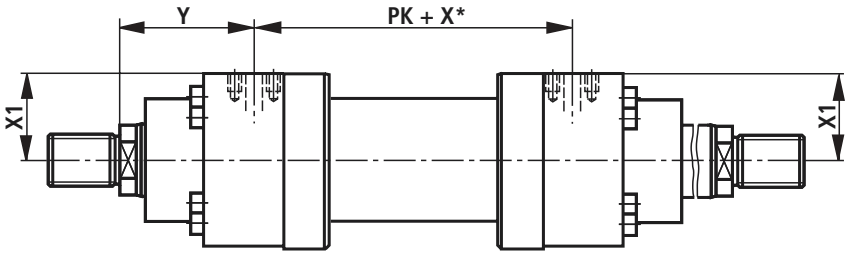
10) = Counter bore 2 mm deep, for S.H.C.S.; DIN 912 – The screws must not be subjected to shear forces. The forces should be distributed via keys.

Flange connections

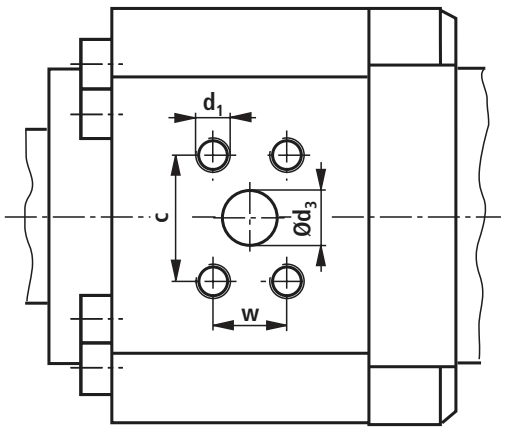
CDH1



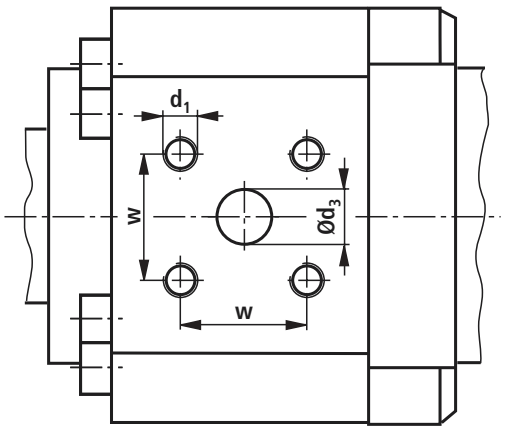
CGH1



Porting pattern for rectangular flanges to ISO 6162 table 2 (≙ SAE 6000 PSI)



Porting pattern for square flanges to ISO 6164 table 2



Flange connections

Dimensions (nominal dimensions in mm)

AL	Version „D“ ISO 6162 Tab.2 (400 bar) (≙ SAE 6000 PSI)											Version „H“ ISO 6164 Tab.2 (400 bar)								
Ø	Y	PJ PK	X1	d_3 Ø	d_3 4) Ø	c ±0.25	w ±0.25	d_1	t_1 1)	t_1 2)	p 3)	Y	PJ PK	X1	d_3 Ø	w ±0.25	d1	t_1 1)	t_1 2)	p 3)
40	–	–	–	–	–	–	–	–	–	–	–	78	122	40.5	10	24.7	M6	12.5	10	400
50	–	–	–	–	–	–	–	–	–	–	–	86	122	48	10	24.7	M6	12.5	10	400
63	–	–	–	–	–	–	–	–	–	–	–	99	135	57	13	29.7	M8	16	13	400
80	102.5	149	65	13	1/2“	40.5	18.2	M8	16	14	400	103	148	67	13	29.7	M8	16	15	400
100	124	171	80.5	13	1/2“	40.5	18.2	M8	16	16	400	123	173	81.5	19	35.4	M8	16	16	400
125	135	205	97.5	19	3/4“	50.8	23.8	M10	20	20	400	131.5	212	99	25	43.8	M10	20	20	400
140	152	227	107	25	1“	57.2	27.8	M12	24	24	400	152	227	109	25	43.8	M10	20	20	400
160	184	242	127	25	1“	57.2	27.8	M12	24	24	400	182.5	245	128	32	51.6	M12	24	24	400
180	199	264	139.5	32	1 1/4“	66.6	31.8	M14	26	26	400	199	264	142	32	51.6	M12	24	24	400
200	205	278	149	32	1 1/4“	66.6	31.8	M14	26	26	400	201.5	285	149.5	38	60.1	M16	30	30	400
220	242	326	168	38	1 1/2“	79.3	36.5	M16	30	30	400	242	326	171	38	60.1	M16	30	30	400
250	266	326	189	38	1 1/2“	79.3	36.5	M16	30	30	400	266	326	192	38	60.1	M16	30	30	400
280	282	375	204	38	1 1/2“	79.3	36.5	M16	30	30	400	282	375	207	38	60.1	M16	30	30	400
320	287	431	236	51	2“	96,8	44,5	M20	36	36	400	287	431	240	51	69,3	M16	30	30	400

For main dimensions see pages 6 to 17

AL = Piston Ø

X* = Stroke length

1) = Thread depth for seal versions M, T and S

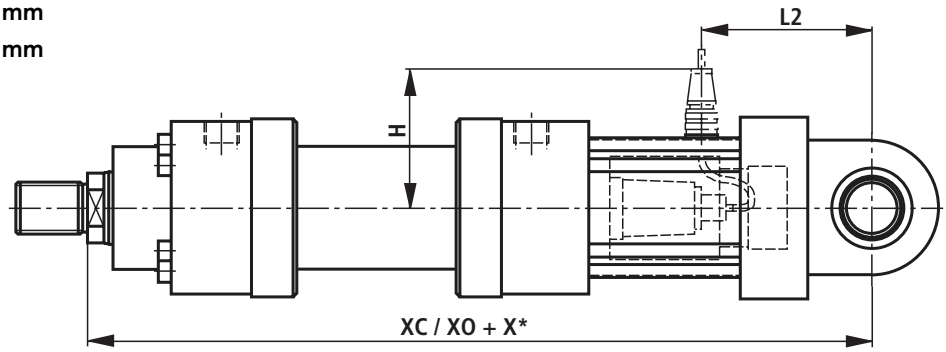
2) = Thread depth for seal versions A and B

3) = Max. operating pressure for associated flanges in bar

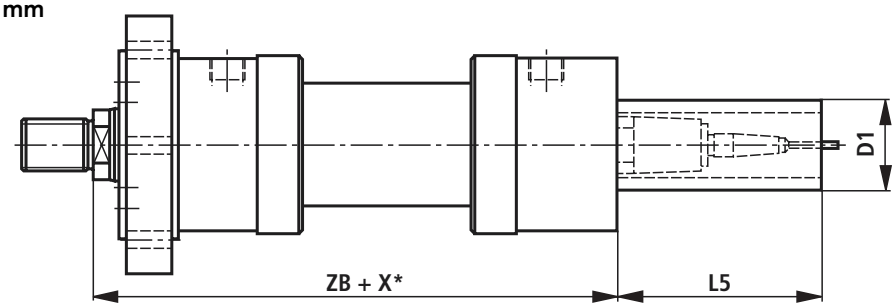
4) = Flange porting pattern to ISO 6162 table 2 relates to a flange porting pattern to SAE 6000 PSI

Position measuring system

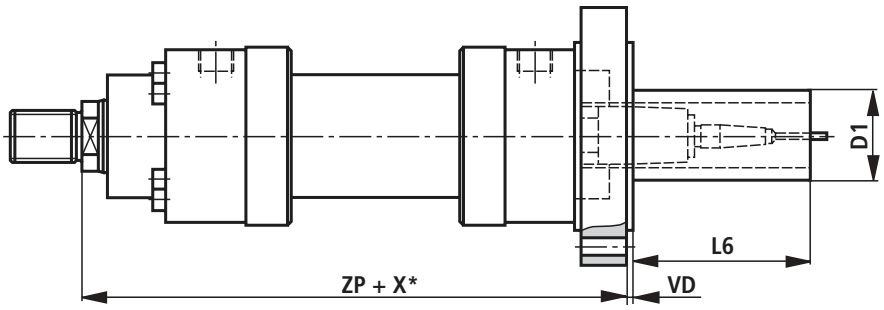
MP3: AL-Ø 40-200 mm
MP5: AL-Ø 40-320 mm



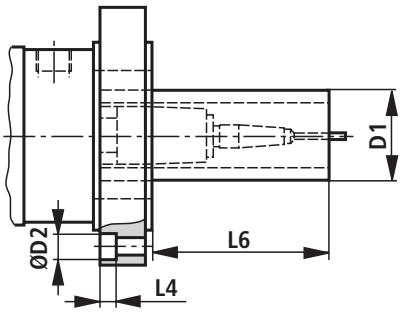
MF3: AL-Ø 40-320 mm



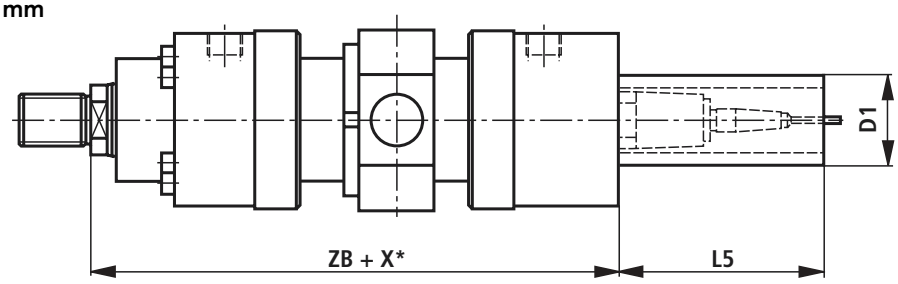
MF4: AL-Ø 40-63 mm



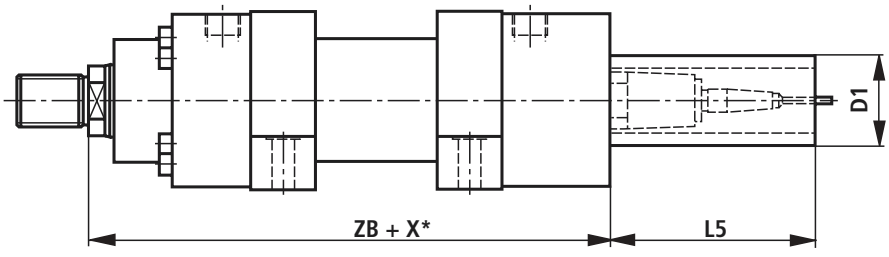
MF4: AL-Ø 80-320 mm



MT4: AL-Ø 40-320 mm



MS2: AL-Ø 40-320 mm



Position measuring system

Dimensions (nominal dimensions in mm)

AL Ø	MM Ø	X^*_{max}	XC	XO	H	ZB	ZP	VD	L2	L4	L5	L6	D1 max	D2 Ø
40	**– 28**	1400	417	417	115	235	265	5	98	0	166	166	80	0
50	**28 36**	1400	430	430	120	243	274	5	103	0	166	166	96	0
63	**36 45**	2000	480	480	130	287	310	5	116	0	166	166	96	0
80	**45 56**	2000	515	515	125	312	330	5	132	0	166	143	96	0
100	**56 70**	3000	560	560	135	352	390	5	145	0	166	123	96	0
125	**70 90**	3000	620	620	145	392	432	5	172	21.5	166	121	96	33
140	**90 100**	3000	665	665	155	430	475	10	182	21.5	166	111	96	33
160	**100 110**	3000	720	720	165	475	535	10	200	28.5	166	96	96	43
180	**110 125**	3000	775	775	175	515	585	10	222	28.5	166	86	96	43
200	**125 140**	3000	815	815	190	535	615	10	237	32	166	76	96	48
220	**140 160**	3000	–	960	205	635	720	10	280	32	166	71	96	48
250	**160 180**	3000	–	1000	220	659	744	10	300	38	166	71	96	57
280	**180 200**	3000	–	1105	270	744	839	10	330	38	166	61	96	57
320	**200 220**	3000	–	1210	300	815	935	10	375	44	166	36	96	66

For the main dimensions see pages 6 to 17

AL = Piston Ø

MM = Piston rod Ø

X* = Stroke length

X^*_{max} = Max. stroke length

Position measuring system

The contactless, absolute position measuring system is rated up to 500 bar. Its function principle is based on the magnetostrictive effect. In connection with this, a torsion impulse is triggered off when two magnetic fields meet. This impulse is directed from the point of measurement via the wave guide inside the measuring scale to the sensor head. The transmission time is constant and virtually independent of temperature. It is propotional to the position of the magnet and can therefore be used as a reference for the actual position value and is converted into a direct analogue or digital output in the sensor head.

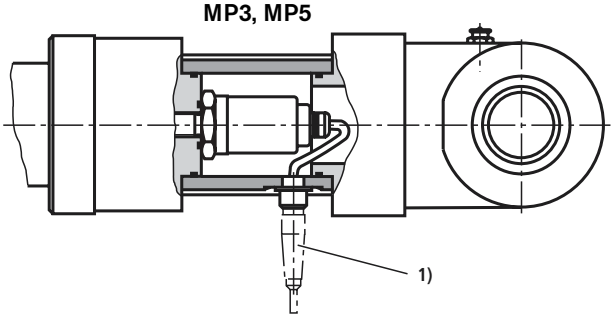
Technical data (for applications outside these parameters, please consult us!)

Operating pressure		bar	250
Analogue output		V	0 to 10
	Load resistance	kΩ	≥ 5
	Resolution		Infinite
Analogue output		mA	4 to 20
	Load resistance	Ω	0 to 500
	Resolution		Infinite
Digital output			SSI 24 Bit grey coded
	Resolution	µm	5
	Measuring direction		Forwards
Linearity (absolute accuracy)	Analogue	% mm	≤ ±0.02 % (with reference to the measurement length) min. ±0,05
	Digital	% mm	≤ ±0.01 % (with reference to the measurement length) min. ±0,04
Repeatability		% mm	±0.001 (with reference to the measurement length) min. ±0.0025
Hysteresis		mm	≤ 0.004
Supply voltage		V DC	24 (± 10 % with an analogue output)
	Current consumption	mA	100
	Residual ripple	% s-s	≤ 1
	Current consumption	V DC mA	24 (+ 20 %/– 15 % with a digital output) 70
	Residual ripple	% s-s	≤ 1
Insulation	Tube and flange		IP 67
	Sensor electronics		IP 65
Operating temperature	Sensor electronics	°C	– 40 to + 75
Temperature co-efficient	Voltage	ppm/°C	70
	Current	ppm/°C	90

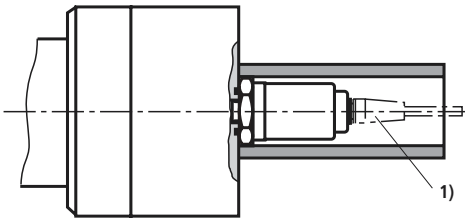
Position measuring system

Mounting style

MP3, MP5



MF3, MF4, MT4, MS2



1) For analogue output:
6-pin Amphenol -
Plug-in connector, Material No. **R900072231**
(plug-in connector is **not** included within the scope of supply, must be ordered separately)

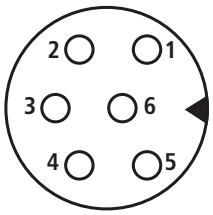


1) For digital output:
7-pin Amphenol -
Plug-in connector, Material No. **R900079551**
(plug-in connector is **not** included within the scope of supply, must be ordered separately)



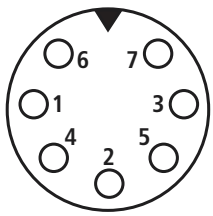
Pin allocation

Position measuring system (analogue output)
Component plug (viewed on the pin side)



Pin	Cable	Signal / current	Signal / voltage
1	grey	4...20 mA	0 - 10 V
2	pink	Gnd	Gnd
3	yellow	n. c.	10 - 0 V
4	green	n. c.	Gnd
5	brown	+24 V DC (±10%)	+24 V DC (±10%)
6	white	Gnd	Gnd

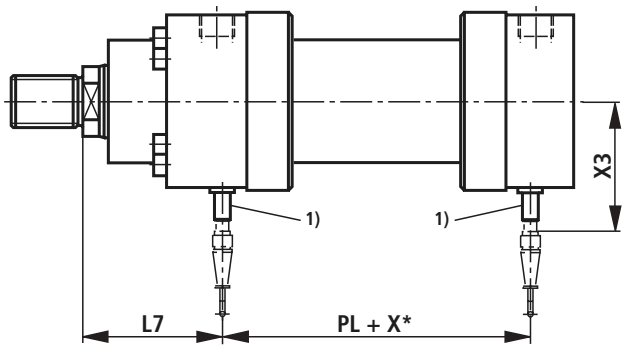
Position measuring system (digital output)
Component plug (viewed on the pin side)



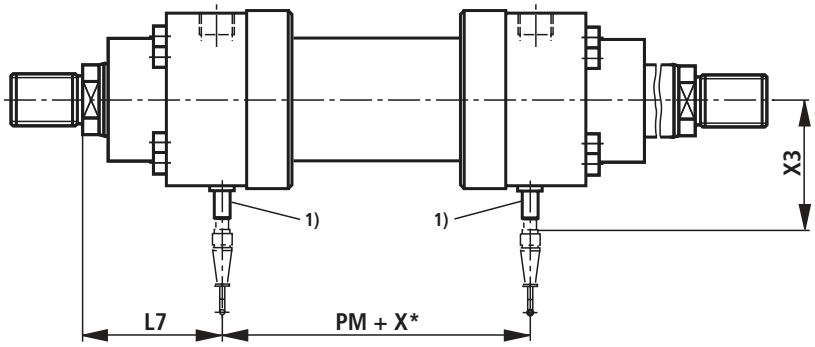
Pin	Cable	Signal / SSi
1	grey	data (–)
2	pink	data (+)
3	yellow	tact (+)
4	green	tact (–)
5	brown	+24 V DC (+20%/–15%)
6	white	0 V
7	–	n. c.

Proximity switch

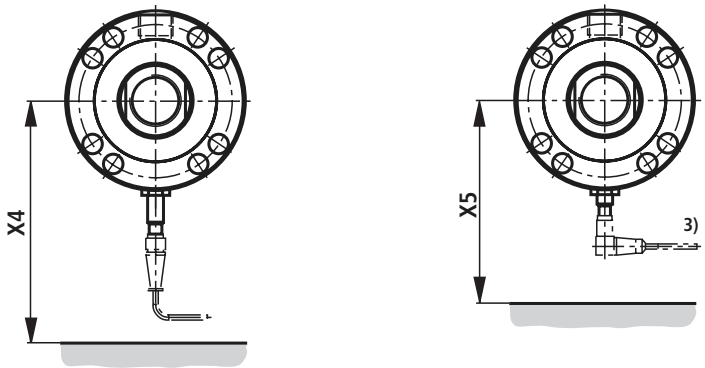
CDH1



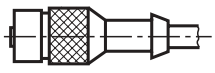
CGH1



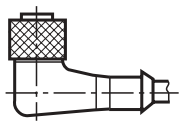
Mounting style



Plug-in connector with a 5 m cable
Material No. **R900026512**
(plug-in connector is **not** included within the scope of supply, must be ordered separately)



Plug-in connector, angled
with a 5 m cable
(the orientation of the cable exit is not definable)
Material No. **R900021404**
(plug-in connector is **not** included within the scope of supply, must be ordered separately)



Proximity switch

Dimensions (nominal dimensions in mm)

AL Ø	MM Ø	PL	PM	L7	X3	X4	X5
40	**22 28**	112	112	83	94	170	125
50	**28 36**	110	110	92	98	175	130
63	**36 45**	125	125	104	103	180	135
80	**45 56**	138	138	108	108	185	140
100	**56 70**	161	161	129	116	195	150
125	**70 90**	189	189	143	126	205	160
140	**90 100**	209	209	161	146	225	180
160	**100 110**	228	228	191	151	230	185
180	**110 125**	254	254	204	159	235	190
200	**125 140**	264	264	212	166	245	200
220	**140 160**	310	310	250	177 [2)]	255	– [3)]
250	**160 180**	310	310	274	187 [2)]	265	– [3)]
280	**180 200**	369	369	285	189 [2)]	275	– [3)]
320	**200 220**	415	415	295	209 [2)]	285	– [3)]

For the main dimensions see pages 6 to 17

AL = Piston Ø

MM = Piston rod Ø

X* = Stroke length

1) = The proximity switch is always located opposite to the pipe connection

2) = Piston Ø 220 - 320 mm
The proximity switch does not overhang

3) = Piston Ø 220 - 320 mm
The angled plug-in connector is not possible

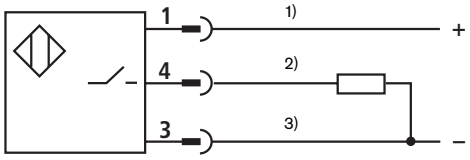
Proximity switch

Inductive proximity switches are used as a reliable end position control for hydraulic cylinders. They are an important element, in providing safety systems, interlocks and/or other machine functions where signals safety and exactly monitor the end position. The up to 500 bar rated high pressure proximity switches operates without contact and contacts and are therefore wear-free. Due to safety reasons the proximity switch is protected against being screwed in too deep. The switching distance can therefore not be adjusted. For the version with proximity switches (Option 1 „E“), the cylinders are fitted with proximity switches on each end.

Technical data (for applications outside these parameters, please consult us!)

Function			PNP N/O
Permissible pressure		bar	500
Operating voltage		V DC	10 … 30
	Including the residual ripple	%	≤ 15
Voltage drop		V	≤ 1.5
Rated operating voltage		V DC	24
Rated operating current		mA	200
No load current		mA	≤ 8
Residual current		µA	≤ 10
Repeatability		%	≤ 5
Hysteresis		%	≤ 15
Ambient temperature range		°C	– 25 … + 80
Temperature drift		%	≤ 10
Switching frequency		Hz	1000
Protection	Active area		IP 68 to DIN 40050
	Proximity switch		IP 67 to DIN 40050
Housing material			Material No. 1.4104

Pin allocation





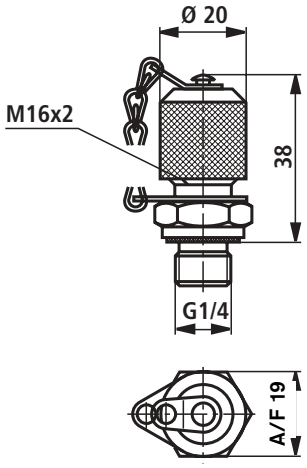
1) Brown
2) Black
3) Blue

Coupling

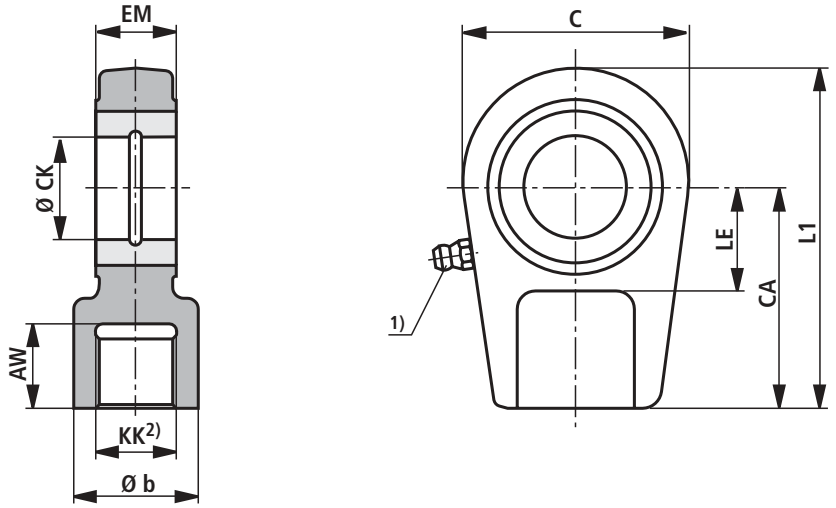
For pressure measurement or bleeding.
For installaion in the bleed/measuring port. Coupling with check valve function, i.e. it can also be connected when pressure is present.

Scope of supply:

Coupling AB 20-11/K1 with NBR seals,
Material No. **R900009090**
Coupling AB 20-11/K1 V with FKM seals,
Material No. **R900001264**



Plain clevis CSA (nominal dimensions in mm)



AL Ø	Typ	Material No.	AW	b	C	CA	CK H11	EM −0.4	KK	LE	L1	*m* 3) kg
40	**CSA 16**	R900303150	17	28	56	50	25	23	M16x1.5	25	80	0.43
50	**CSA 22**	R900303151	23	34	64	60	30	28	M22x1.5	30	94	0.7
63	**CSA 28**	R900303152	29	44	78	70	35	30	M28x1.5	40	112	1.1
80	**CSA 35**	R900303153	36	55	94	85	40	35	M35x1.5	45	135	2.0
100	**CSA 45**	R900303154	46	70	116	105	50	40	M45x1.5	55	168	3.3
125	**CSA 58**	R900303155	59	87	130	130	60	60	M58x1.5	65	200	5.5
140	**CSA 65**	R900303156	66	93	154	150	70	55	M65x1.5	75	232	8.6
160	**CSA 80**	R900303157	81	125	176	170	80	60	M80x2	80	265	12.2
180	**CSA100**	R900303158	101	143	206	210	90	65	M100x2	90	323	21.5
200	**CSA110**	R900303159	111	153	230	235	100	70	M110x2	105	360	27.5

AL = Piston Ø

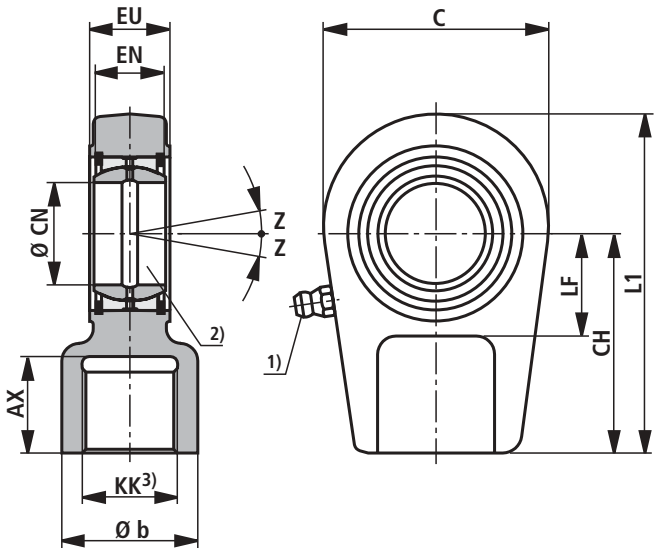
1) = Grease nipple; cone head form A to DIN 71412

2) = The plain clevis must always be screwed on up to the piston rod thread stop

3) *m* = Weight of the plain clevis

Self-aligning clevis CGA (nominal dimensions in mm)

AL-Ø 40-280 mm



AL Ø	Type	Material No.	AX	b	C	CH	CN	EN	EU –0.4	KK	L1	LF	Z	*m* 4) kg
40	**CGA 16**	R900303125	17	28	56	50	$25_{-0.010}$	$20_{-0.12}$	23	M16x1.5	80	25	8°	0.43
50	**CGA 22**	R900303126	23	34	64	60	$30_{-0.010}$	$22_{-0.12}$	28	M22x1.5	94	30	7°	0.7
63	**CGA 28**	R900303127	29	44	78	70	$35_{-0.012}$	$25_{-0.12}$	30	M28x1.5	112	40	7°	1.1
80	**CGA 35**	R900303128	36	55	94	85	$40_{-0.012}$	$28_{-0.12}$	35	M35x1.5	135	45	7°	2.0
100	**CGA 45**	R900303129	46	70	116	105	$50_{-0.012}$	$35_{-0.12}$	40	M45x1.5	168	55	7°	3.3
125	**CGA 58**	R900303130	59	87	130	130	$60_{-0.015}$	$44_{-0.15}$	50	M58x1.5	200	65	7°	5.5
140	**CGA 65**	R900303131	66	93	154	150	$70_{-0.015}$	$49_{-0.15}$	55	M65x1.5	232	75	6°	8.6
160	**CGA 80**	R900303132	81	125	176	170	$80_{-0.015}$	$55_{-0.15}$	60	M80x2	265	80	6°	12.2
180	**CGA100**	R900303133	101	143	206	210	$90_{-0.020}$	$60_{-0.20}$	65	M100x2	323	90	6°	21.5
200	**CGA110**	R900303134	111	153	230	235	$100_{-0.020}$	$70_{-0.20}$	70	M110x2	360	105	7°	27.5
220	**CGA120**	R900303135	125	176	265	265	$110_{-0.020}$	$70_{-0.20}$	80	M120x3	407,5	115	6°	40.7
250	**CGA120**	R900303135	125	176	265	265	$110_{-0.020}$	$70_{-0.20}$	80	M120x3	407,5	115	6°	40.7
280	**CGA130**	R900303136	135	188	340	310	$120_{-0.020}$	$85_{-0.20}$	90	M130x3	490	140	6°	76.4
320	–	–	–	–	–	–	–	–	–	–	–	–	–	–

AL = Piston Ø

1) = Grease nipple; cone head form A to DIN 71412

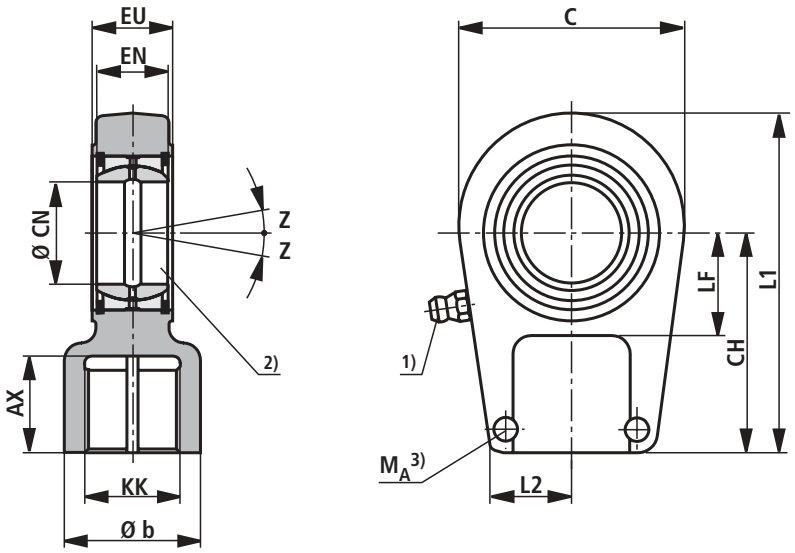
2) = Associated pin Ø m6;
associated pin Ø j6 with a maintenance-free self-aligning clevis

3) = The self-aligning clevis must always be screwed on up to the piston rod thread stop

4) *m* = Weight of the self-aligning clevis

Self-aligning clevis CGAK (nominal dimensions in mm)

AL-Ø 40-280 mm



AL Ø	Type	Material No.	AX	b	C	CH	CN	EN	EU −0.4	KK	L1	L2	LF	M_A 3) Nm	Z	m 4) kg
40	**CGAK 16**	R900303162	17	28	56	50	$25_{-0.010}$	$20_{-0.12}$	23	M16x1.5	80	20	25	9	8°	0.43
50	**CGAK 22**	R900303163	23	34	64	60	$30_{-0.010}$	$22_{-0.12}$	28	M22x1.5	94	22	30	20	7°	0.7
63	**CGAK 28**	R900303164	29	44	78	70	$35_{-0.012}$	$25_{-0.12}$	30	M28x1.5	112	27	40	20	7°	1.1
80	**CGAK 35**	R900303165	36	55	94	85	$40_{-0.012}$	$28_{-0.12}$	35	M35x1.5	135	35	45	40	7°	2.0
100	**CGAK 45**	R900303166	46	70	116	105	$50_{-0.012}$	$35_{-0.12}$	40	M45x1.5	168	42	55	80	7°	3.3
125	**CGAK 58**	R900303167	59	87	130	130	$60_{-0.015}$	$44_{-0.15}$	50	M58x1.5	200	54	65	160	7°	5.5
140	**CGAK 65**	R900303168	66	93	154	150	$70_{-0.015}$	$49_{-0.15}$	55	M65x1.5	232	57	75	160	6°	8.6
160	**CGAK 80**	R900303169	81	125	176	170	$80_{-0.015}$	$55_{-0.15}$	60	M80x2	265	66	80	160	6°	12.2
180	**CGAK100**	R900321655	101	143	206	210	$90_{-0.020}$	$60_{-0.20}$	65	M100x2	323	76	90	160	6°	21.5
200	**CGAK110**	R900321691	111	153	230	235	$100_{-0.020}$	$70_{-0.20}$	70	M110x2	360	85	105	300	7°	27.5
220	**CGAK120**	R900321621	125	176	265	265	$110_{-0.020}$	$70_{-0.20}$	80	M120x3	407,5	96	115	500	6°	40.7
250	**CGAK120**	R900321621	125	176	265	265	$110_{-0.020}$	$70_{-0.20}$	80	M120x3	407,5	96	115	500	6°	40.7
280	**CGAK130**	R900322015	135	188	340	310	$120_{-0.020}$	$85_{-0.20}$	90	M130x3	490	112	140	500	6°	76.4
320	**–**	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–

AL = Piston Ø

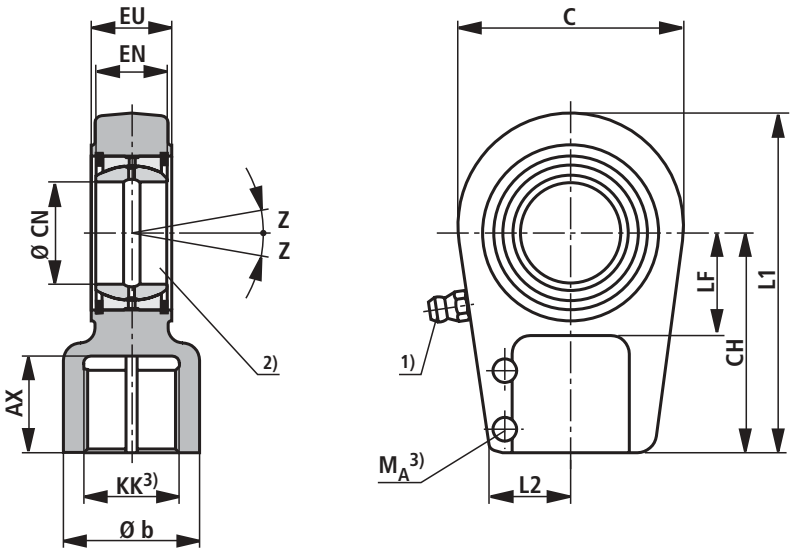
1) = Grease nipple; cone head form A to DIN 71412

2) = Associated pin Ø m6;
associated pin Ø j6 with a maintenance-free self-aligning clevis

3) M_A = Tightening torque
The self-aligning clevis must always be screwed on up to the piston rod thread stop. Subsequently, the clamping screws have to be tightned to the specified torque.

4) m = Weight of the self-aligning clevis

Self-aligning clevis CGAS (nominal dimensions in mm)



AL Ø	Type	Material No.	AX	b	C	CH	CN	EN	EU −0.4	KK	L1	L2	LF	M_A 3) Nm	Z	m 4) ca. kg
40	**CGAS 25**	R900303137	30	28	56	65	$25_{-0.010}$	$20_{-0.12}$	23	M18x2	95	24	25	20	8°	0,65
50	**CGAS 30**	R900303138	35	34	64	75	$30_{-0.010}$	$22_{-0.12}$	28	M24x2	109	28	30	20	7°	1,0
63	**CGAS 35**	R900303139	46	44	78	90	$35_{-0.012}$	$25_{-0.12}$	30	M30x2	132	35	40	40	7°	1,5
80	**CGAS 40**	R900303140	56	55	94	105	$40_{-0.012}$	$28_{-0.12}$	35	M39x3	155	39	45	80	7°	2,4
100	**CGAS 50**	R900303141	76	70	116	135	$50_{-0.012}$	$35_{-0.12}$	40	M50x3	198	45	55	80	7°	4,8
125	**CGAS 60**	R900303142	96	87	130	170	$60_{-0.015}$	$44_{-0.15}$	50	M64x3	240	59	65	160	7°	8,6
140	**CGAS 70**	R900303143	112	105	154	195	$70_{-0.015}$	$49_{-0.15}$	55	M80x3	278	69	75	160	6°	12,2
160	**CGAS 80**	R900303144	122	125	176	210	$80_{-0.015}$	$55_{-0.15}$	60	M90x3	305	84	80	300	6°	18,4
180	**CGAS 90**	R900303145	142	150	206	250	$90_{-0.020}$	$60_{-0.20}$	65	M100x3	363	90	90	300	5°	31,6
200	**CGAS100**	R900303146	152	170	230	275	$100_{-0.020}$	$70_{-0.20}$	70	M110x4	400	94	105	300	7°	34
220	**CGAS110**	R900303147	162	180	264	300	$110_{-0.020}$	$70_{-0.20}$	80	M120x4	442	105	115	500	6°	44
250	**CGAS110**	R900303147	162	180	264	300	$110_{-0.020}$	$70_{-0.20}$	80	M120x4	442	105	115	500	6°	44
280	**CGAS120**	R900303148	192	210	340	360	$120_{-0.020}$	$85_{-0.20}$	90	M150x4	540	120	140	500	6°	75
320	**CGAS140**	R900317314	210	230	380	420	$140_{-0.025}$	$90_{-0.25}$	110	M160x4	620	128	185	1000	7°	160

AL = Piston Ø

1) = Grease nipple; cone head form A to DIN 71412

2) = Associated pin Ø m6;
associated pin Ø j6 with a maintenance-free self-aligning clevis

3) M_A = Tightening torque
The self-aligning clevis must always be screwed on up to the piston rod thread stop. Subsequently, the clamping screws have to be tightened to the specified torque.

4) m = Weight of the self-aligning clevis

Buckling

The permissible stroke length with a flexibly guided load and a 3.5 safety factor against buckling can be obtained from the appropriate table. When a deviating cylinder installation orientation the permissible stroke length has to be interpolated. Permissible stroke lengths for non-guided loads are available on request.

The calculations for buckling are carried out as follows:

1. Calculation according to Euler

$$F = \frac{\pi^2 \cdot E \cdot I}{\nu \cdot L_K^2} \quad \text{if } \lambda > \lambda g$$

2. Calculation according to Tetmajer

$$F = \frac{d^2 \cdot \pi \,(335\text{-}0{,}62 \cdot \lambda)}{4 \cdot \nu} \quad \text{if } \lambda \le \lambda g$$

Explanation:

E = Modulus of elasticity in N/mm²
= 2.1 x 10^5 for steel

I = Moment of inertia in mm⁴ for a cicular cross-section $= \frac{d^4 \cdot \pi}{64} = 0.0491 \cdot d^4$

ν = 3.5 (safety factor)

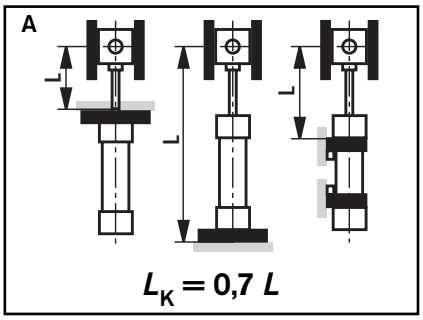
L_K = Free buckling length in mm (dependent on the mounting style, see sketches A, B, C)

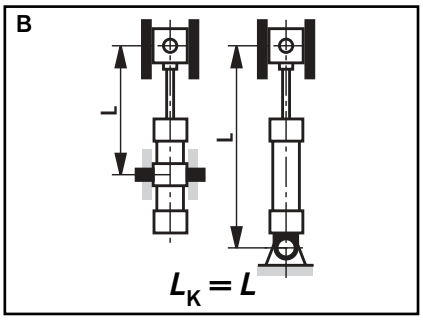
d = Piston rod Ø in mm

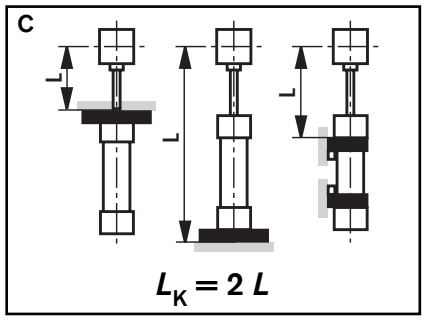
λ = Slenderness ratio $= \frac{4 \cdot L_K}{d}$ $\quad \lambda g = \pi \sqrt{\frac{E}{0.8 \cdot R_e}}$

R_e = Yield strength of the piston rod material

The influence of the mounting style on the buckling length:







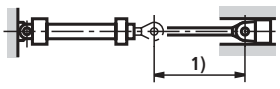
Permissible stroke lengths (nominal dimensions in mm)

Mounting style CDH1: MP3, MP5

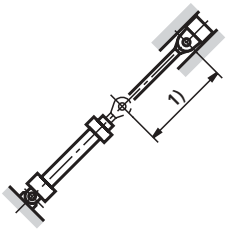
AL Ø	MM Ø	Permissible stroke lengths at 100 bar 0°	100 bar 45°	100 bar 90°	160 bar 0°	160 bar 45°	160 bar 90°	250 bar 0°	250 bar 45°	250 bar 90°	Max. available stroke lengths
40	22	195	200	215	130	135	140	40	45	55	2000
	28	385	400	445	295	300	320	215	220	225	
50	28	285	295	310	205	210	215	120	130	135	
	36	535	555	625	425	430	460	320	325	335	
63	36	390	400	440	290	295	305	200	205	210	
	45	655	685	790	530	545	585	410	415	430	
80	45	500	515	560	375	385	400	240	260	280	
	56	815	850	980	665	680	735	520	525	545	
100	56	610	630	705	470	480	505	280	295	355	3000
	70	985	1030	1240	820	845	930	650	660	695	
125	70	770	800	900	600	615	650	360	380	465	
	90	1295	1360	1670	1095	1130	1265	885	900	955	
140	90	1145	1200	1430	945	970	1070	740	755	790	
	100	1400	1475	1840	1190	1230	1390	965	985	1050	
160	100	1230	1285	1530	1010	1040	1140	790	800	840	
	110	1480	1555	1930	1250	1290	1455	1005	1030	1090	
180	110	1305	1365	1630	1065	1095	1200	825	840	880	
	125	1675	1765	2210	1420	1470	1670	1150	1175	1260	
200	125	1500	1580	1930	1240	1290	1430	985	1005	1060	
	140	1865	1965	2520	1590	1660	1910	1305	1340	1440	
220	140	1620	1710	2180	1360	1415	1630	1090	1120	1200	6000
	160	2075	2200	3000	1810	1890	2280	1510	1560	1730	
250	160	1885	1990	2570	1600	1670	1930	1300	1330	1440	
	180	2330	2475	3370	2040	2135	2570	1710	1770	1960	
280	180	2075	2200	2900	1775	1880	2170	1450	1490	1620	
	200	2510	2670	3700	2200	2310	2820	1850	1920	2140	
320	200	2170	2300	3070	1850	1940	2290	1500	1550	1700	
	220	2590	2760	3850	2260	2380	2920	1890	1960	2200	

Installation

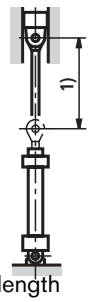
0°



45°



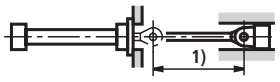
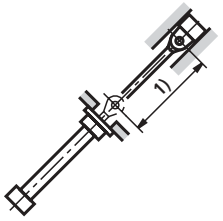
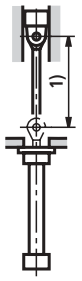
90°



1) Perm. stroke length

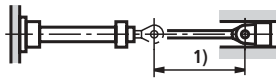
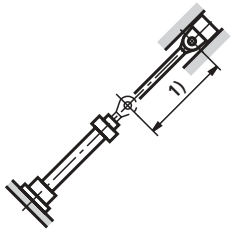
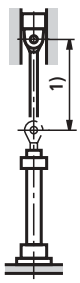
Permissible stroke lengths (nominal dimensions in mm)

Mounting style CDH1: MF3

AL Ø	MM Ø	Permissible stroke lengths at 100 bar 0°	100 bar 45°	100 bar 90°	160 bar 0°	160 bar 45°	160 bar 90°	250 bar 0°	250 bar 45°	250 bar 90°	Max. available stroke lengths	Installation
40	22	895	915	980	730	735	760	440	450	510	2000	0° 
	28	1400	1415	1630	1180	1205	1275	970	980	1010		
50	28	1180	1200	1280	955	965	995	700	730	780		
	36	1785	1855	2160	1530	1570	1695	1275	1290	1340		
63	36	1520	1560	1690	1250	1270	1315	1010	1015	1035		
	45	2000	2000	2000	1875	1925	2000	1570	1595	1670		
80	45	1855	1905	2000	1540	1560	1630	1140	1180	1280		
	56	2000	2000	2000	2000	2000	2000	1910	1940	2000		
100	56	2250	2320	2500	1880	1910	2010	1300	1360	1580	3000	45° 
	70	3000	3000	3000	2770	2860	3000	2360	2400	2550		
125	70	2760	2860	3000	2330	2375	2520	1580	1680	1990		
	90	3000	3000	3000	3000	3000	3000	3000	3000	3000		
140	90	3000	3000	3000	3000	3000	3000	2770	2820	2980		
	100	3000	3000	3000	3000	3000	3000	3000	3000	3000		
160	100	3000	3000	3000	3000	3000	3000	2980	3000	3000		
	110	3000	3000	3000	3000	3000	3000	3000	3000	3000		
180	110	3000	3000	3000	3000	3000	3000	3000	3000	3000		
	125	3000	3000	3000	3000	3000	3000	3000	3000	3000		
200	125	3000	3000	3000	3000	3000	3000	3000	3000	3000		
	140	3000	3000	3000	3000	3000	3000	3000	3000	3000		
220	140	5400	5680	6000	4800	4980	5780	4120	4220	4560	6000	90° 
	160	6000	6000	6000	5820	6000	6000	5150	5330	6000		
250	160	6000	6000	6000	5450	5660	6000	4720	4840	5290		
	180	6000	6000	6000	6000	6000	6000	5730	5920	6000		
280	180	6000	6000	6000	6000	6000	6000	5270	5420	5970		
	200	6000	6000	6000	6000	6000	6000	6000	6000	6000		
320	200	6000	6000	6000	6000	6000	6000	6000	6000	6000		
	220	6000	6000	6000	6000	6000	6000	6000	6000	6000		

[1] Perm. stroke length

Mounting style CDH1: MF4

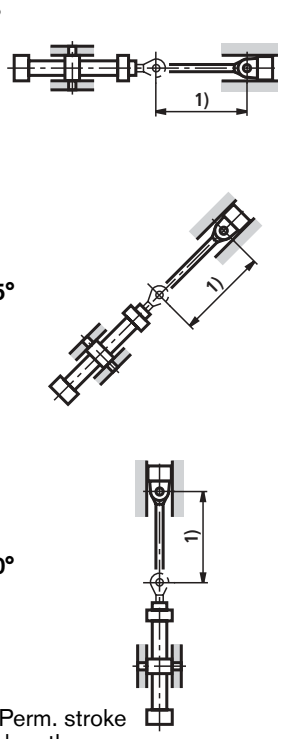
AL Ø	MM Ø	Permissible stroke lengths at 100 bar 0°	100 bar 45°	100 bar 90°	160 bar 0°	160 bar 45°	160 bar 90°	250 bar 0°	250 bar 45°	250 bar 90°	Max. available stroke lengths	Installation
40	22	325	340	370	245	250	260	105	110	140	2000	0° 
	28	565	590	695	465	475	520	365	370	385		
50	28	455	470	515	350	360	375	220	230	265		
	36	770	805	960	640	660	725	515	525	550		
63	36	600	620	710	475	490	520	350	370	380		
	45	930	975	1210	790	820	920	645	660	700		
80	45	760	785	895	610	625	670	395	420	495		
	56	1150	1210	1495	985	1020	1145	810	825	875		
100	56	905	945	1120	745	765	835	420	460	620	3000	45° 
	70	1370	1445	1880	1190	1235	1440	995	1020	1100		
125	70	1175	1225	1460	980	1010	1105	580	620	835		
	90	1815	1920	2560	1600	1670	1980	1365	1400	1540		
140	90	1600	1695	2190	1390	1440	1670	1150	1180	1275		
	100	1915	2030	2770	1695	1770	2130	1440	1490	1650		
160	100	1730	1825	2350	1490	1550	1790	1235	1265	1365		
	110	2030	2155	2910	1790	1870	2240	1520	1565	1720		
180	110	1850	1950	2510	1590	1655	1900	1310	1340	1450		
	125	2295	2440	3000	2030	2130	2570	1730	1785	1980		
200	125	2110	2230	2270	1835	1910	2250	1530	1575	1720		
	140	2540	2700	3000	2265	2380	2930	1945	2010	2260		
220	140	2250	2400	3350	1990	2090	2550	1685	1740	1950	6000	90° 
	160	2800	2990	4500	2530	2680	3480	2220	2310	2700		
250	160	2615	2780	3900	2320	2435	3000	1980	2050	2300		
	180	3140	3360	5050	2850	3010	3910	2500	2610	3050		
280	180	2850	3050	4400	2550	2680	3370	2190	2270	2600		
	200	3370	3610	5550	3070	3250	4300	2700	2820	3330		
320	200	3000	3210	4700	2680	2830	3590	2100	2390	2750		
	220	3500	3750	5800	3180	3370	4480	2790	2920	3460		

[1] Perm. stroke length

Permissible stroke lengths (nominal dimensions in mm)

Mounting style CDH1: MT4 trunnions in the middle of the cylinder

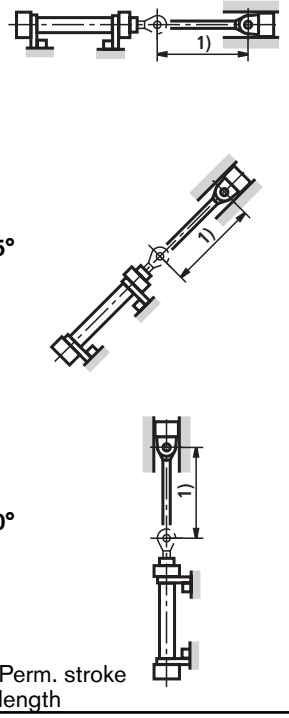
AL Ø	MM Ø	Permissible stroke lengths at 100 bar 0°	100 bar 45°	100 bar 90°	160 bar 0°	160 bar 45°	160 bar 90°	250 bar 0°	250 bar 45°	250 bar 90°	Max. available stroke lengths	Installation
40	22	340	345	365	250	255	260	130	135	145	2000	0°
	28	590	605	665	470	480	500	365	370	375		
50	28	460	470	495	350	355	365	245	250	260		
	36	790	815	910	645	655	690	510	515	525		
63	36	610	625	675	475	485	500	360	365	370		
	45	965	1000	1140	800	815	870	635	645	665		
80	45	770	790	850	605	615	635	440	455	475		
	56	1190	1235	1410	990	1010	1080	795	805	830		
100	56	930	955	1060	745	755	795	490	510	595	3000	45°
	70	1430	1490	1770	1210	1240	1360	985	1000	1045		
125	70	1185	1225	1360	960	980	1030	640	670	780		
	90	1885	1970	2390	1620	1665	1850	1340	1360	1430		
140	90	1675	1710	2060	1410	1415	1575	1140	1155	1205		
	100	2020	2115	2610	1735	1790	2010	1440	1465	1555		
160	100	1805	1880	2210	1510	1550	1680	1215	1230	1285		
	110	2140	2240	2740	1830	1885	2100	1505	1535	1620		
180	110	1925	2005	2360	1605	1650	1790	1290	1310	1360		
	125	2420	2540	3000	2080	2150	2420	1720	1755	1865		
200	125	2130	2230	2690	1790	1840	2040	1440	1465	1540		
	140	2610	2750	3000	2250	2330	2670	1865	1910	2050		
220	140	2490	2510	3150	2050	2120	2400	1685	1720	1835	6000	90°
	160	3000	3170	4230	2640	2750	3260	2240	2310	2530		
250	160	2750	2900	3660	2380	2460	2810	1970	2020	2160		
	180	3350	3540	4750	2960	3090	3670	2520	2600	2850		
280	180	3040	3210	4140	2640	2750	3170	2210	2260	2440		
	200	3620	3840	5210	3210	3360	4040	2750	2830	3140		
320	200	3210	3390	4410	2790	2900	3380	2320	2380	2580		
	220	3770	4000	5450	3340	3490	4200	2850	2930	3250		



1) Perm. stroke length

Mounting style CDH1: MS2

AL Ø	MM Ø	Permissible stroke lengths at 100 bar 0°	100 bar 45°	100 bar 90°	160 bar 0°	160 bar 45°	160 bar 90°	250 bar 0°	250 bar 45°	250 bar 90°	Max. available stroke lengths	Installation
40	22	825	840	885	645	650	665	370	375	410	2000	0°
	28	1305	1350	1535	1085	1110	1180	875	885	910		
50	28	1075	1100	1175	855	865	890	610	625	675		
	36	1680	1750	2000	1430	1465	1590	1175	1190	1240		
63	36	1405	1440	1570	1135	1155	1200	895	900	920		
	45	2000	2000	2000	1760	1810	1990	1460	1480	1555		
80	45	1730	1780	1960	1410	1435	1500	1000	1050	1155		
	56	2000	2000	2000	2000	2000	2000	1785	1820	1920		
100	56	2110	2180	2440	1740	1770	1870	1140	1220	1440	3000	45°
	70	3000	3000	3000	2620	2710	3000	2210	2260	2400		
125	70	2600	2695	3000	2170	2210	2360	1400	1480	1820		
	90	3000	3000	3000	3000	3000	3000	2890	2970	3000		
140	90	3000	3000	3000	3000	3000	3000	2585	2635	2800		
	100	3000	3000	3000	3000	3000	3000	3000	3000	3000		
160	100	3000	3000	3000	3000	3000	3000	2760	2810	2990		
	110	3000	3000	3000	3000	3000	3000	3000	3000	3000		
180	110	3000	3000	3000	3000	3000	3000	2940	3000	3000		
	125	3000	3000	3000	3000	3000	3000	3000	3000	3000		
200	125	3000	3000	3000	3000	3000	3000	3000	3000	3000		
	140	3000	3000	3000	3000	3000	3000	3000	3000	3000		
220	140	5090	5370	6000	4490	4670	5470	3820	3910	4260	6000	90°
	160	6000	6000	6000	5510	5800	6000	4850	5020	5750		
250	160	5790	6000	6000	5150	5370	6000	4420	4540	4990		
	180	6000	6000	6000	6000	6000	6000	5420	5630	6000		
280	180	6000	6000	6000	5700	5960	6000	4930	5070	5630		
	200	6000	6000	6000	6000	6000	6000	6000	6000	6000		
320	200	6000	6000	6000	6000	6000	6000	5200	5400	6000		
	220	6000	6000	6000	6000	6000	6000	6000	6000	6000		



1) Perm. stroke length

End position cushioning

End position cushioning:
The objective is to reduce the speed of a moving mass, whose centre of gravity lies on the cylinder axis, to a level, at which neither the cylinder nor the machine, into which the cylinder is installed, can be damaged. For speeds over 20 mm/s we recommend the use of end position cushioning so that the energy can be absorbed without the use of any additional equipment. Checks must always be carried out whether, also at lower speeds, with high masses end position cushioning is required.

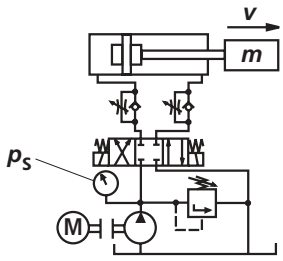
Cushioning capacity:
When braking masses via the end position cushioning, the design contricted cushioning capacity must not be exceeded. Cylinders with end position cushioning can only utilise their full cushioning capacity when the entire stroke length is used.

With the adjustable end position cushioning „E“ an additional throttle valve is added to the „D“ version. The end position cushioning „E“ makes it possible to optimise the cycle times. The maximum cushioning capacity can only be achieved when the throttle valve is fully closed.

The calculation depends on the factors of weight, velocity, system pressure and installation position. Therefore, the variable D_m is to be calculated from weight and speed, the variable D_p from system pressure and installation position. These variables are then used to verify the permissible cushioning performance in the „cushioning capacity“ diagram. The interseption point of the variables D_m and D_p must always be below the cushioning capacity curve of the selected cylinder. The values in the diagram refer to an average oil temperature of + 45 to +65 °C and with a closed throttle valve.

For special applications with very short stroke times, high speeds or masses, special cylinders with special end position cushioning can be offered on request.

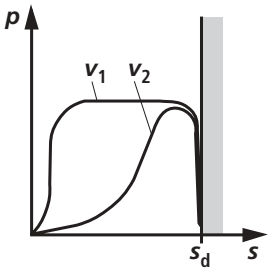
When fixed or adjustable end stops are used, then special measures have to be taken!



Formulas:

$$D_m = \frac{m}{10^K};\ K = kv\ (0.5–v)$$

m = Moved weight in kg
v = Stroke velocity in m/s
kv = See table on page 35



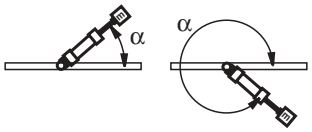
Extending:

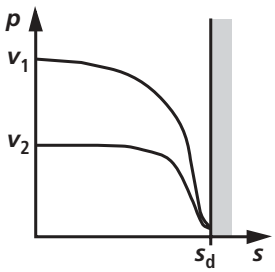
$$D_p = p_S - \frac{m \cdot 9.81 \cdot \sin\alpha}{A_1 \cdot 10}$$

Retracting:

$$D_p = p_S + \frac{m \cdot 9.81 \cdot \sin\alpha}{A_3 \cdot 10}$$

p_S = System pressure in bar
A_1 = Piston area in cm² (see page 3)
A_3 = Annulus in cm² (see page 3)
α = Angle in degrees with reference to the horizontal plane





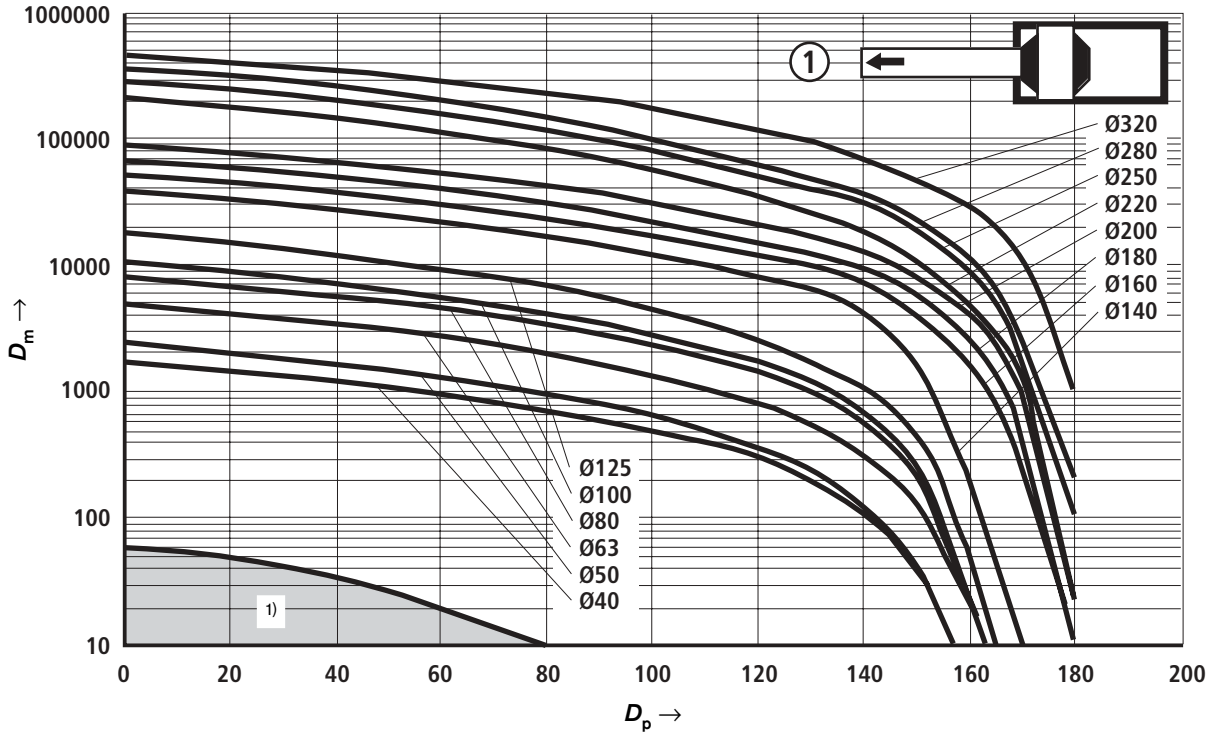
Damping length

AL Ø mm	40	50	63	80	100	125	140	160	180	200	220	250	280	320
Head side	21	20	23	25	25	25	33	33	37	37	76	81	86	90
Base side	21	20	23	25	25	25	33	33	37	37	76	81	86	90

End position cushioning

AL Ø mm	40	50	63	80	100	125	140	160	180	200	220	250	280	320
kv ①	2.85	2.97	2.56	2.82	3.51	3.02	2.53	2.65	2.91	2.76	2.85	2.95	3.11	3.13
kv ②	3.1	3.25	2.85	2.85	3.52	2.91	2.53	2.93	2.95	2.95	2.93	3.1	3.12	3.07
kv ③	2.95	3.1	2.73	3.1	3.51	2.95	2.51	2.91	2.95	2.91	2.93	2.93	3.15	3.25

Cushioning capacity: Extending

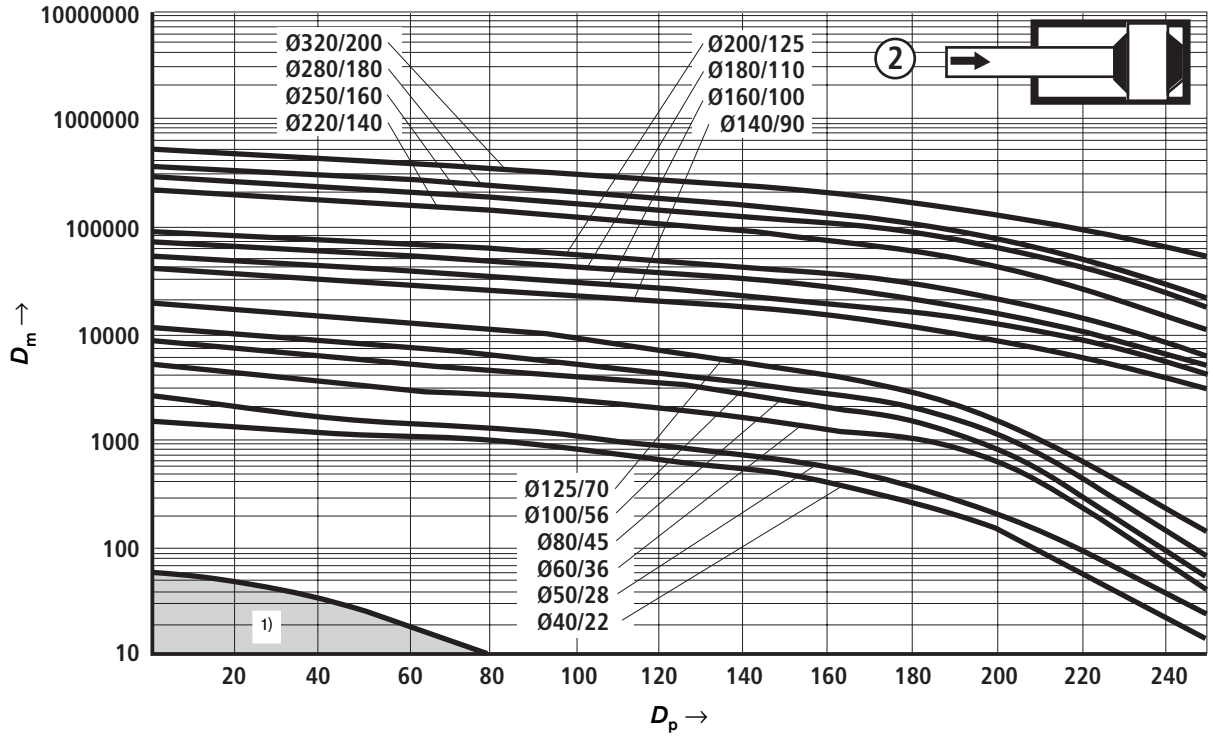


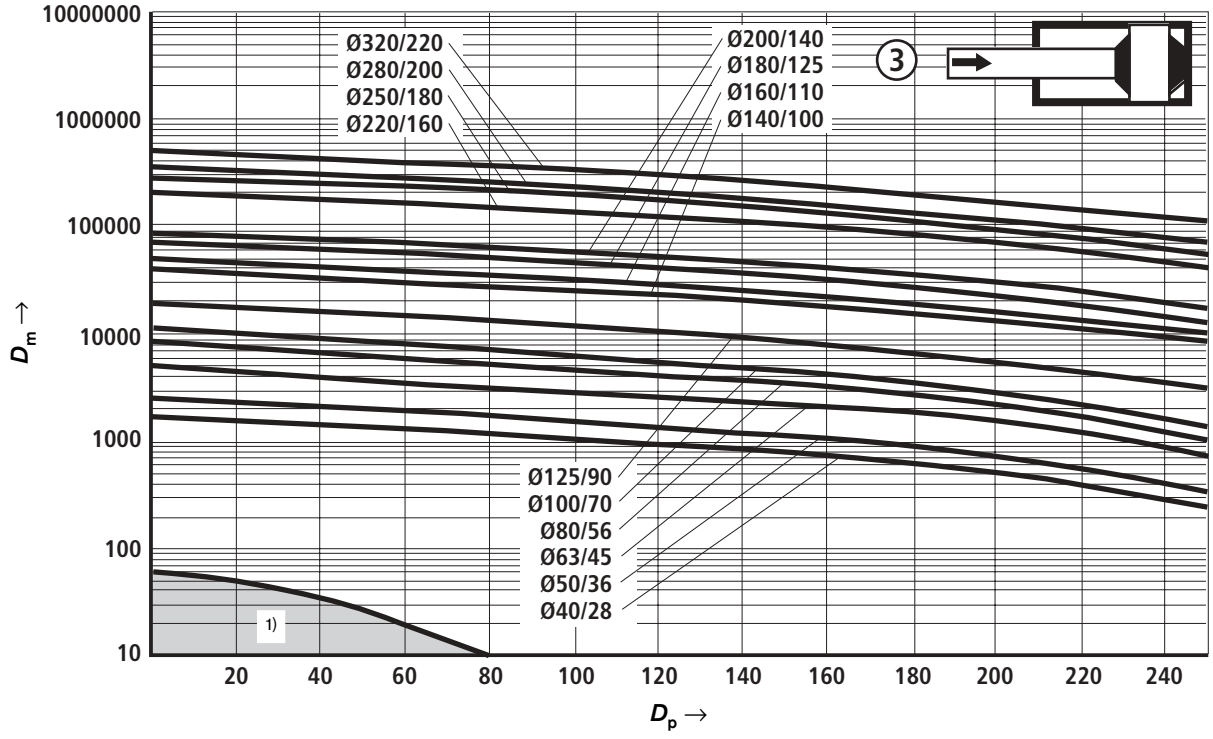
AL = Piston Ø

1) If, for standard applications, the calculated interception point from D_m and D_p is within the indicated area, then we recommend that a cylinder is used without end position damping.

End position cushioning

Cushioning capacity: Retracting

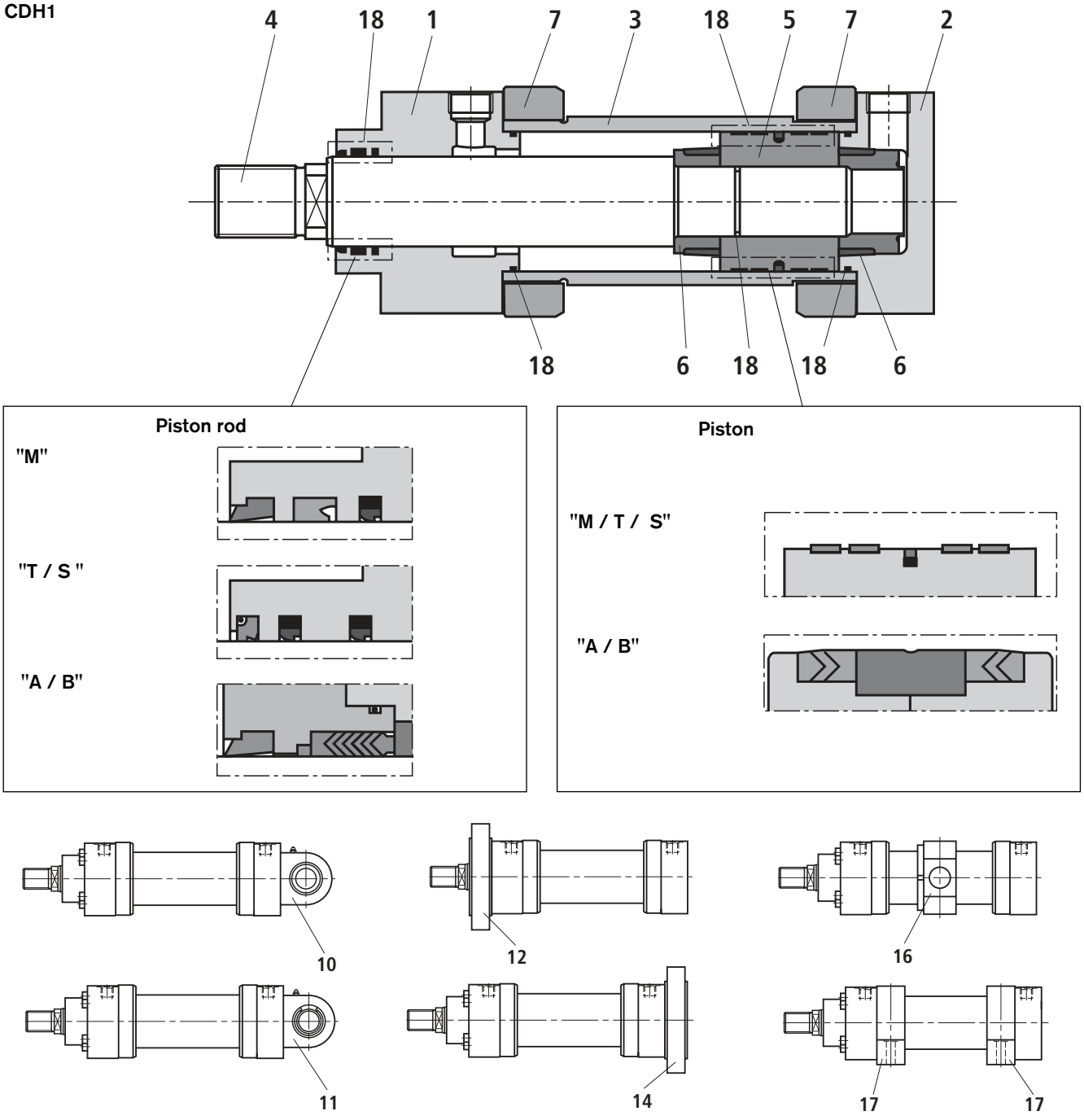




[1] If, for standard applications, the calculated interception point from D_m and D_p is within the indicated area, then we recommend that a cylinder is used without end position damping.

Spare parts

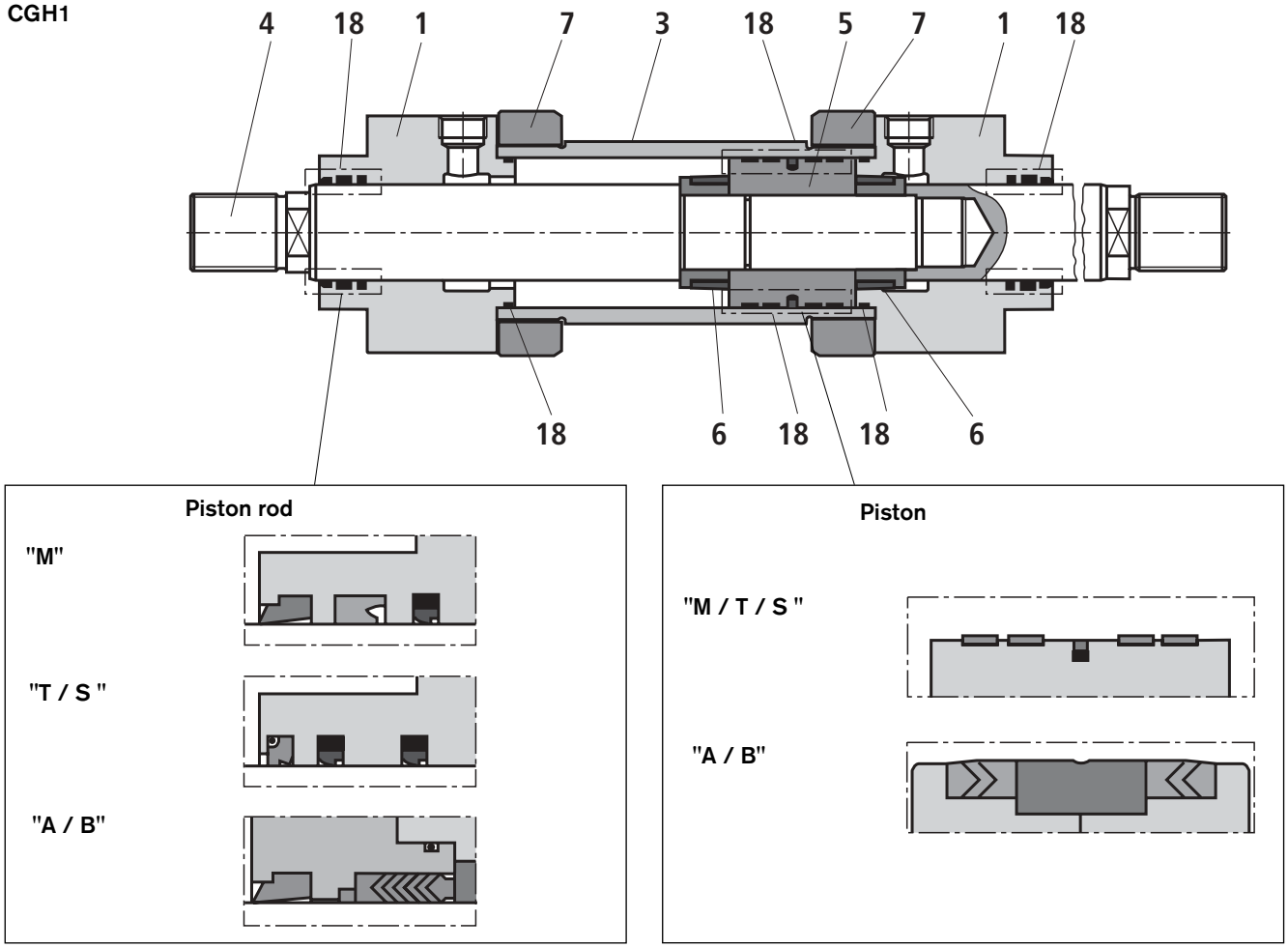
CDH1

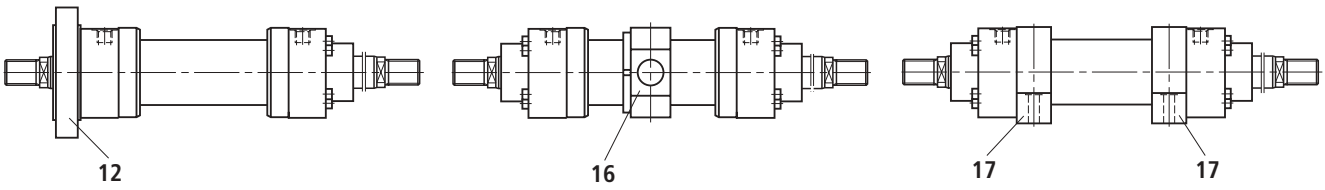


1 Head
2 Base
3 Barrel
4 Piston rod
5 Piston
6 Damping bush
7 Flange
10 Base MP3
11 Base MP5
12 Round flange MF3
14 Round flange MF4
16 Trunnion MT4
17 Foot MS2
18 Seal kit :
Wiper
Rod seal
Piston seal
O-ring
Guide ring

Spare parts

CGH1

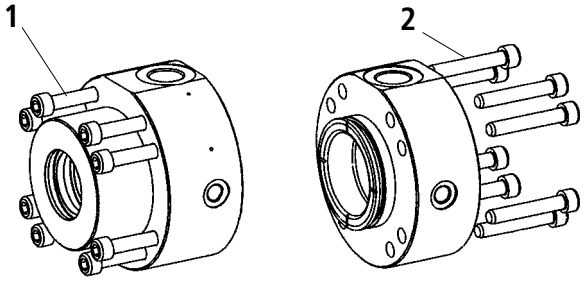




1 Head
3 Barrel
4 Piston rod
5 Piston
6 Damping bush
7 Flange
12 Round flange MF3
16 Trunnion MT4
17 Foot MS2
18 Seal kit:
Wiper
Rod seal
Piston seal
O-ring
Guide ring

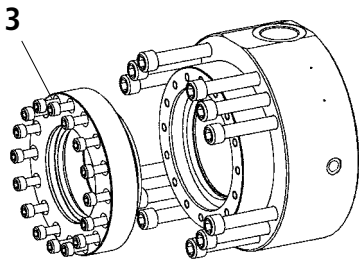
Tightening torque

Screws: Head and base (Pos. 1 and 2)



Series	Piston Ø	Screw	Quantity	Grade	Tightening torque
CDH1 / CGH1	**40**	M8	4	10.9	23 Nm
CDH1 / CGH1	**50**	M8	8	10.9	20 Nm
CDH1 / CGH1	**63**	M8	8	10.9	30 Nm
CDH1 / CGH1	**80**	M10	8	10.9	55 Nm
CDH1 / CGH1	**100**	M12	8	10.9	100 Nm
CDH1 / CGH1	**125**	M16	8	10.9	200 Nm
CDH1 / CGH1	**140**	M16	12	10.9	170 Nm
CDH1 / CGH1	**160**	M16	12	10.9	220 Nm
CDH1 / CGH1	**180**	M20	12	10.9	350 Nm
CDH1 / CGH1	**200**	M20	12	10.9	410 Nm
CDH1 / CGH1	**220**	M20	16	10.9	460 Nm
CDH1 / CGH1	**250**	M24	16	10.9	700 Nm
CDH1 / CGH1	**280**	M24	16	10.9	800 Nm
CDH1 / CGH1	**320**	M30	16	10.9	1500 Nm

Screws: Seal cover (Pos. 3)



Only with seal versions "A" and "B"

Series	Piston Ø	Piston rod Ø	Screw	Quantity	Grade	Tightening torque
CDH1 / CGH1	**160**	**100**	M10	16	10.9	60 Nm
		110				
CDH1 / CGH1	**180**	**110**	M12	16	10.9	80 Nm
		125				
CDH1 / CGH1	**200**	**125**	M12	16	10.9	90 Nm
		140				
CDH1 / CGH1	**220**	**140**	M12	16	10.9	90 Nm
		160		24		
CDH1 / CGH1	**250**	**160**	M12	24	10.9	90 Nm
		180				
CDH1 / CGH1	**280**	**180**	M12	24	10.9	90 Nm
		200				
CDH1 / CGH1	**320**	**200**	M12	24	10.9	90 Nm
		220	M16	16		230 Nm

Seal kits[1)]

CDH1 – Standard

AL	MM	Material No. for seal version				
Ø	Ø	M	T	A	S	B
40	22	R900850072	R900849536	R900860270	R900861000	R900859816
	28	R900851087	R900858841	R900859445	R900861001	R900859770
50	28	R900850181	R900857535	R900860928	R900861002	R900860938
	36	R900849392	R900860277	R900851515	R900861004	R900860940
63	36	R900850191	R900860278	R900860930	R900861005	R900851206
	45	R900847956	R900847855	R900851638	R900861007	R900859678
80	45	R900851086	R900860280	R900854708	R900861008	R900860942
	56	R900850905	R900856180	R900854718	R900861010	R900851205
100	56	R900853936	R900860282	R900860470	R900861011	R900860944
	70	R900853382	R900860285	R900856094	R900861013	R900860946
125	70	R900853966	R900860286	R900854709	R900861014	R900860948
	90	R900857949	R900856102	R900856095	R900861016	R900855464
140	90	R900858281	R900860289	R900860932	R900861017	R900860951
	100	R900853965	R900860290	R900856096	R900849080	R900860952
160	100	R900855683	R900860291	R900860468	R900861018	R900860953
	110	R900851146	R900857536	R900860933	R900861019	R900860954
180	110	R900856497	R900852561	R900860934	R900861020	R900860955
	125	R900848603	R900860292	R900860935	R900861021	R900860956
200	125	R900860294	R900860295	R900860936	R900861022	R900860957
	140	R900856431	R900860293	R900860937	R900861023	R900860958
220	140	R900888100	R900888108	R900888116	R900888132	R900888140
	160	R900888101	R900888109	R900888117	R900888133	R900888141
250	160	R900888102	R900888110	R900888118	R900888134	R900888142
	180	R900888103	R900888111	R900888119	R900888135	R900888143
280	180	R900888104	R900888112	R900888120	R900888136	R900888144
	200	R900888105	R900888113	R900888121	R900888137	R900888145
320	200	R900888106	R900888114	R900888122	R900888138	R900888146
	220	R900888107	R900888115	R900888123	R900888139	R900888147

AL = Piston Ø in mm

MM = Piston rod Ø in mm

1) = Seal kits for position measuring systems and proximity switches have separate Material Nos.

Seal kits[1)]

CGH1 – Standard

AL	MM	Material No. for seal version				
Ø	Ø	M	T	A	S	B
40	22	R900867251	R900868888	R900866746	R900868942	R900867132
	28	R900867252	R900868889	R900866747	R900868943	R900867133
50	28	R900867253	R900868890	R900866748	R900868944	R900867134
	36	R900864930	R900868892	R900866750	R900868946	R900867136
63	36	R900867260	R900868893	R900866751	R900868947	R900867137
	45	R900867262	R900868895	R900866753	R900868949	R900867139
80	45	R900867263	R900868896	R900866754	R900868950	R900867140
	56	R900867265	R900868898	R900866756	R900868952	R900867142
100	56	R900867266	R900868899	R900866757	R900868953	R900867143
	70	R900867268	R900868901	R900866759	R900868955	R900867146
125	70	R900867269	R900868902	R900866760	R900867906	R900867147
	90	R900867270	R900868904	R900866762	R900868957	R900867149
140	90	R900867271	R900868905	R900866763	R900868958	R900867150
	100	R900867272	R900868906	R900866764	R900868959	R900867151
160	100	R900867273	R900868907	R900866765	R900868960	R900867152
	110	R900867274	R900868908	R900866766	R900868961	R900867153
180	110	R900867275	R900868909	R900866767	R900868962	R900867154
	125	R900867276	R900868910	R900866768	R900868963	R900867155
200	125	R900867277	R900868911	R900866769	R900868964	R900867156
	140	R900867278	R900868912	R900866770	R900868965	R900867157
220	140	R900888020	R900888028	R900888036	R900888052	R900888060
	160	R900888021	R900888029	R900888037	R900888053	R900888061
250	160	R900888022	R900888030	R900888038	R900888054	R900888062
	180	R900888023	R900888031	R900888039	R900888055	R900888063
280	180	R900888024	R900888032	R900888040	R900888056	R900888064
	200	R900888025	R900888033	R900888041	R900888057	R900888065
320	200	R900888026	R900888034	R900888042	R900888058	R900888066
	220	R900888027	R900888035	R900888043	R900888059	R900888067

AL = Piston Ø

MM = Piston rod Ø

1) = Seal kits for proximity switches have separate Material Nos.

Seal kits[1)]

CDH1 – Standard + Option F

AL Ø	MM Ø	Material No. for seal version		
		M+F	T+F	S+F
40	22	R900861024	R900861049	R900861099
	28	R900861025	R900861050	R900861100
50	28	R900861026	R900861051	R900861101
	36	R900861028	R900861053	R900861103
63	36	R900861029	R900861054	R900861104
	45	R900861031	R900861056	R900861106
80	45	R900861032	R900861057	R900861107
	56	R900861034	R900861059	R900861109
100	56	R900861035	R900861060	R900861112
	70	R900861037	R900861062	R900861115
125	70	R900861038	R900861063	R900861117
	90	R900861040	R900861065	R900861122
140	90	R900861041	R900861066	R900861124
	100	R900861042	R900861067	R900861126
160	100	R900861043	R900861068	R900861128
	110	R900861044	R900861069	R900861130
180	110	R900861045	R900861070	R900861133
	125	R900861046	R900861071	R900861135
200	125	R900861047	R900861072	R900861142
	140	R900861048	R900861073	R900861143

CGH1 – Standard + Option F

AL Ø	MM Ø	Material No. for seal version		
		M+F	T+F	S+F
40	22	R900868998	R900869025	R900869092
	28	R900868999	R900869026	R900869093
50	28	R900869000	R900869027	R900869094
	36	R900869002	R900869029	R900869096
63	36	R900869003	R900869030	R900869097
	45	R900869005	R900869032	R900869099
80	45	R900869006	R900869033	R900869100
	56	R900869008	R900869035	R900869102
100	56	R900869009	R900869036	R900869103
	70	R900869013	R900869038	R900869105
125	70	R900869014	R900869039	R900869106
	90	R900869016	R900869041	R900869108
140	90	R900869017	R900869042	R900869109
	100	R900869018	R900869043	R900869110
160	100	R900869019	R900869044	R900869111
	110	R900869020	R900869045	R900869112
180	110	R900869021	R900869046	R900869113
	125	R900869022	R900869047	R900869114
200	125	R900869023	R900869048	R900869115
	140	R900869024	R900869049	R900869116

AL = Piston Ø

MM = Piston rod Ø

[1)] = Seal kits for proximity switches have separate Material Nos.

Seal kits

Only for position measuring system

AL	Material No. for seal version				
Ø	M / M+F	T / T+F	A	S / S+F	B
40	R900885935		–	R900885937	–
50	R900894958		–	R900894979	–
63	R900894959		–	R900894980	–
80	R900894960		–	R900894981	–
100	R900894961		–	R900894982	–
125	R900894962		–	R900894983	–
140	R900894963		–	R900894985	–
160	R900894964		–	R900894986	–
180	R900894973		–	R900894987	–
200	R900894974		–	R900894988	–
220	R900894975		–	R900894989	–
250	R900894976		–	R900894991	–
280	R900894977		–	R900894993	–
320	R900894978		–	R900894994	–

Only for proximity switch

AL	Material No. for seal version				
Ø	M / M+F	T / T+F	A	S / S+F	B
40 to 200	R900885938			R900885939	
220 to 320	R900894997			R900894998	

AL = Piston Ø

Notes

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Bosch Rexroth AG
Hydraulics
Zum Eisengießer 1
97816 Lohr am Main, Germany
Telefon +49 (0) 93 52 / 18-0
Telefax +49 (0) 93 52 / 18-23 58
documentation@boschrexroth.de
www.boschrexroth.de

Bosch Rexroth Teknik AB
Varuvägen 7, Älvsjö
S-125 81 Stockholm
Telefon +46 (08) 72 79 20 0
Telefax +46 (08) 86 87 21
cyl.hyd@boschrexroth.se
www.boschrexroth.se

Bosch Rexroth SA
BP 37 - Z.I. Les Fourmis
F-74131 Bonneville Cedex
Telefon +33 (0) 4 50 25 35 45
Telefax +33 (0) 4 50 25 35 19
www.boschrexroth.fr